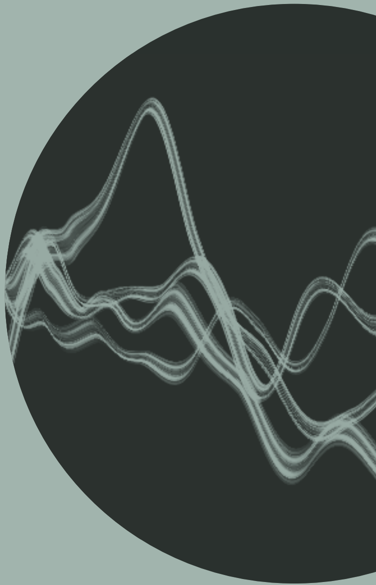
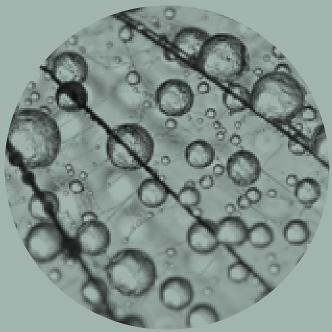
CHARTING THE FUTURE OF INNOVATION | VOLUME 100 | 2019–02

ERICSSON TECHNOLOGY Review



TECHNOLOGY TRENDS
MANIFESTING THE INNOVATION PLATFORM

BLOCKCHAINS
FACILITATING TRUST ONLINE



CLOUD-NATIVE
APPLICATION DESIGN IN THE TELECOM DOMAIN

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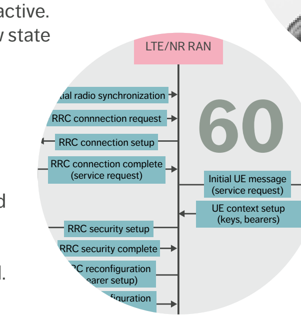
LTE/NR RAN
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RRC connection request
RRC connection setup
RRC connection complete (service request)
Initial UE message (service request)
UE context setup (keys, bearers)
RRC security setup
RRC security complete

Ericsson Technology Review brings you insights into some of the key emerging innovations that are shaping the future of ICT. Our aim is to encourage an open discussion about the potential, practicalities, and benefits of a wide range of technical developments, and provide insight into what the future has to offer.

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FEATURE ARTICLE
Six key trends manifesting the platform for innovation by Erik Ekudden

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THE RISE OF THE INNOVATION PLATFORM

■ **THERE'S NO DOUBT ABOUT IT:** society and industry are transforming at an unprecedented rate in response to new technologies in areas such as the IoT, distributed computing and AI, and connectivity is playing a pivotal role. Self-driving vehicles, intelligent manufacturing robots and real-time drone control are just a few examples.

The trends I highlighted in 2018 as the five to watch were right on target, and they have only continued to grow in strength and relevance over the course of the past year. In this year's trends article, which you can find on page 28, I build on last year's conclusions and share my view of the future network platform in relation to an updated list that now includes six trends. The evolution characterized by this year's trends points to 5G and beyond, toward the future definition of 6G.

I truly believe that the defining characteristic of the future network platform will be its ability to instantaneously meet any application need, anytime. Achieving this requires ubiquitous radio access, security assurance, zero-touch networks, and distributed compute and storage – four of this year's six trends. The other two trends – the Internet of Skills and cyber-physical systems – are important examples of use cases that a future network platform needs to support.

The other articles in this issue of the magazine address critical issues such as trust enablement, the extension of computing resources all the way to the edge of the mobile network, the growing impact of the cloud in the telco domain, overcoming latency and battery consumption challenges, and the need for end-to-end connectivity.

> THE EVOLUTION CHARACTERIZED BY THIS YEAR'S TRENDS POINTS TO 5G AND BEYOND

At Ericsson, we see significant value in blockchains as a trust enabler and potential disruptor that can enable completely new business models in the digital asset market. A decade after its launch, blockchain technology is still one of very few internet-age technologies that can facilitate trust online. In this issue, we explore its potential in telco.

Service exposure and APIs will play a key role in creating solutions that enable operators to provide computing resources across the whole telco domain to the edge of the mobile network – a capability that is essential to meet the requirements of use cases in areas such as the IoT, AR/VR, Industry 4.0 and the automotive sector.

The transformation of virtual network functions into cloud-native applications (CNAs) is already underway, and we are determined to make it as smooth as possible. We've developed an application development framework that includes a set of architecture principles, design rules, and best practices that guide the fundamental design decisions for all our CNAs.

As the IoT continues to expand, latency and battery consumption issues are a growing challenge. The new 'inactive state' in the standalone version of the 5G NR standard overcomes those challenges, and increases overall system capacity by decreasing the processing effort in the network.

We know that future industrial automation will be highly dependent on operators' ability to provide deterministic connectivity end to end, and Time-Sensitive Networking is quickly becoming the standard Ethernet-based technology for converged networks of Industry 4.0. Our TSN article explores the benefits of combining TSN features with 5G URLLC capabilities.

I believe that the contents of this issue demonstrate that the network platform has the potential to offer all the connectivity, processing, storage and security needed by current and future applications. Please feel free to share it with your colleagues and business partners. You can find both PDF and HTML versions of it at: www.ericsson.com/ericsson-technology-review

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BLOCKCHAINS AND ONLINE TRUST ✱

FACILITATING ONLINE TRUST WITH blockchains

A decade after its launch, blockchain is still the only internet-age technology that is able to facilitate online trust using mathematics and collective protocolling exclusively.

DANIEL BERGSTRÖM, BEN SMEETS, MIKAEL JAATINEN, JAMES KEMPF, JONAS LUNDBERG, NICKLAS SANDGREN, GASPAR WOSA

One of the fundamental challenges in the online, digital world is that implicit, fundamental concepts in the off-line, physical world need to be formalized and made explicit. Trust is a prime example.

■ In the physical world, trust is intangible but it is nonetheless central to our interactions with other people and to our consumption of services. Creating an online environment in which people feel secure when interacting and consuming in a similar way requires the development of technologies and protocols that formalize and digitalize trust.

The current solution to the challenge of facilitating trust online is to rely on trusted third parties such as banks and major internet companies to act as trust anchors, creating and attesting certificates for people or web-based services. Each device, browser and operating system comes preconfigured with a list of these trusted third parties and their certificates – their digital fingerprints. By instructing our devices to trust the root certificate of the trusted third party, they are able to computationally infer trust in all underlying entities.

The primary weakness of this hierarchical approach to establishing trust stems from the underlying structure of centralized power. The root keys of each certificate authority are a core asset of today's internet, but they are privately managed and sensitive to exposure. Blockchain was originally designed to uproot this hierarchy and create a new kind of trust system for electronic transactions. In essence, the blockchain itself becomes its own trust anchor based on a distributed, transparent and community-driven infrastructure.

A blockchain removes the need for trusted third parties, distributes the centralized power of the certificate authorities, and allows anonymous members to join and contribute to the infrastructure at their own discretion – although at a very high cost in terms of throughput. While digital currencies are strongly associated with blockchains – the "coins" are generated by contributing resources to the networks and spent by making transactions that are processed by the networks – the value of blockchains goes beyond digital currencies.

Public versus private blockchains

Bitcoin and Ethereum are both classified as public, permissionless blockchains. These systems have three properties that form the basis of trust. Firstly, anyone can become a participant by contributing computing resources – there is no need to have a prior relation to any other node in the system. Secondly, generating a new block on the blockchain is computationally expensive, as the consensus mechanism is designed to require a certain amount of wall-clock time to complete regardless of the size of the network. And lastly, it is impossible to predict which contributor will be the first to complete the next block.

If more than half of the computational resources in the system are technically well-behaved, their results will dominate any malicious or malfunctioning nodes that may try to alter the history of the system in an erroneous direction. In the consensus method used in these systems, known as proof of work (PoW), there are no shortcuts to generating new blocks; it can only be done through a computationally intensive hashing process. Other schemes for consensus are being developed and discussed, but these have yet to see widespread use.

The difference between public blockchains and private, permissioned ones is that the latter employ strong identities, user management and a protected data structure. Private blockchains target use cases somewhere between a public blockchain in an untrusted public environment and a distributed database hosted in a fully trusted internal deployment. This segment includes bank consortia, for example, that have a mutual reliance and at least some level of preestablished trust, but where a privately managed backend for transaction management is not a feasible alternative. Due to the difference in network constitution and the presence of at least partial trust, the computationally expensive PoW scheme is not required in private blockchains. Instead, they can use the same consensus algorithms that are used in other distributed systems, designed to compensate for both malicious and malfunctioning nodes.

> “[PRIVATE BLOCKCHAINS] EMPLOY STRONG IDENTITIES, USER MANAGEMENT AND A PROTECTED DATA STRUCTURE”

The differences in scope between public and private blockchains have a large impact on technology choices. From a technical standpoint, there is virtually no overlap between the two different types of blockchains. It is also significant to note that public blockchains are by design very difficult for companies to monetize, which is why most firms have chosen to focus on private blockchains instead.

Terms and abbreviations

ABI – Application Binary Interface | **IoT** – Internet of Things | **JSON** – JavaScript Object Notation | **PoW** – Proof of Work | **REST**– Representational State Transfer | **SOFIE** – Secure Open Federation for Internet Everywhere | **TEE** – Trusted Execution Environment

BLOCKCHAINS AND ONLINE TRUST ✱

The most commonly used software technology to realize private blockchain installations is Hyperledger Fabric.

Key technical properties of suitable use cases

We have identified four key technical properties of the partial-trust use cases that we expect to be suitable for blockchains: (1) a shared trusted history, (2) structure built on multiple stakeholders of equal standing, (3) largely independent nodes, and (4) access to data history.

Shared trusted history

The key benefit of the blockchain is trust between stakeholders, and to establish a history of transactions that is very hard to tamper with.

Multiple, equal stakeholders

The main niche of blockchains lies in the area of partial trust between roughly equal stakeholders.

Largely independent nodes

Use cases where each node operates independently and uses the blockchain for support are desirable due to the relatively high cost and/or delay of running transactions on the blockchain.

Access to data history

Because the historical data is normally retained indefinitely, it is highly beneficial if there is a value to the use case in having access to historical transactions.

> THE SMART CONTRACT WILL MONITOR, VERIFY AND ENFORCE AGREED CONDITIONS AUTOMATICALLY

Related technologies

The technical development and broadening of blockchains is constantly ongoing. By altering or extending the core functionality, we can both widen the scope and applicability of blockchains as a technology and mitigate the limitations of existing offerings.

Smart contracts

With traditional databases, it is straightforward to create software that monitors a database, determines whether or not a certain condition has been fulfilled, and updates the database accordingly. This is exactly what smart contracts do as well, but in the trusted environment of blockchains. A smart contract is neither smart nor a legal contract; rather, it is an agreement between two or more parties that is formulated and enforced with immutable cryptographic code. This code is executed on every node within the blockchain network and determines how data in the distributed ledger is modified. If a smart contract depends on external information, an oracle must be used to feed this information into the ledger to make it accessible to the smart contracts.

Smart contracts remove reliance on trusted intermediaries when making business agreements. Typically, a smart contract includes terms and conditions, performance metrics and possibly penalties. During execution, the smart contract will monitor, verify and enforce agreed conditions automatically, which can potentially save time and money for the parties involved.

The technology behind smart contracts is promising, but there are some caveats; smart contracts need to be very carefully designed and implemented to ensure that the resulting contract acts exactly as intended given any input or event. Misconfigured smart contracts are virtually impossible to cancel (unless they have been designed for renegotiation from the start), which considerably increases the demands of deploying a smart contract.

Hashgraphs

The drawbacks of the PoW consensus algorithm used by public blockchains (in terms of delay, throughput, energy efficiency and transaction costs) have inspired the development of other technologies targeting the challenge of distributed trust. Hashgraphs are one such example. Hashgraphs reorganize the transaction blocks from a chain of blocks to a directed acyclic graph of blocks, which enables new blocks to be added to the system without waiting for all previous blocks to be organized.

The organization of blocks enables multiple lines of transactions to be run in parallel, and in theory allows for a system that has considerably lower delays and higher throughput compared with a conventional blockchain. Hashgraphs also try to replace the computationally expensive PoW consensus algorithms with other approaches to increase the throughput and energy efficiency of the system. Smart contracts can run on hashgraphs in a way that is similar to how they run on blockchains.

Hashgraphs represent a bold technological leap that strives to overcome all the drawbacks of public blockchains. However, current hashgraph technologies are not open and available in the same way as public blockchain technologies are, which arguably makes them better suited to solve different use cases that are closer to those of private blockchains. Some hashgraph technologies are also designed around patented algorithms and built-in claims to parts of the revenue, which goes against the original intention of blockchain to create a decentralized and democratic infrastructure.

Trusted Execution Environments

A Trusted Execution Environment (TEE) is established within an individual device by using an enclave – a hardware-protected part of the CPU chipset that operates on encrypted memory and storage for security purposes. This approach enables the execution of selected software in isolation from the underlying operating system layers, effectively in isolation from any attacks originating from hacking or exploiting operating system software. The technology was initially launched for some chipsets in the early 2000s but has only recently reached wide-scale deployment in device, desktop and server hardware.

> TEEs MAY OFFER A BREAKTHROUGH IN TERMS OF CONSENSUS ALGORITHMS

From a public blockchain perspective, TEEs may offer a breakthrough in terms of consensus algorithms. A key feature of modern TEEs is the ability to attest the code running inside the enclave through a hardware-supported asymmetric key exchange. The ability to execute trusted and verifiable code on otherwise compromised systems lays the foundation for a new generation of consensus algorithms, anchoring the trust in the signature of the code being executed rather than in the work being carried out or the identity of the node owner. Early results of this development in public blockchains show considerably increased transaction speeds and reduced energy consumption. The implications are yet to be fully determined for private blockchains that rely on classical distributed system algorithms for consensus.

Use cases and applications

At Ericsson, we believe that a robust blockchain foundation can increase ecosystem involvement and enable new business models for revenue generation. In light of this, we have been testing the application of blockchain technology in the realm of telecommunication for some time, and we have identified three use cases that are particularly promising in terms of services with monetization potential. One is called the smart contract platform, the second is known as ID brokering, and the third is a Nubo-based virtual services marketplace.

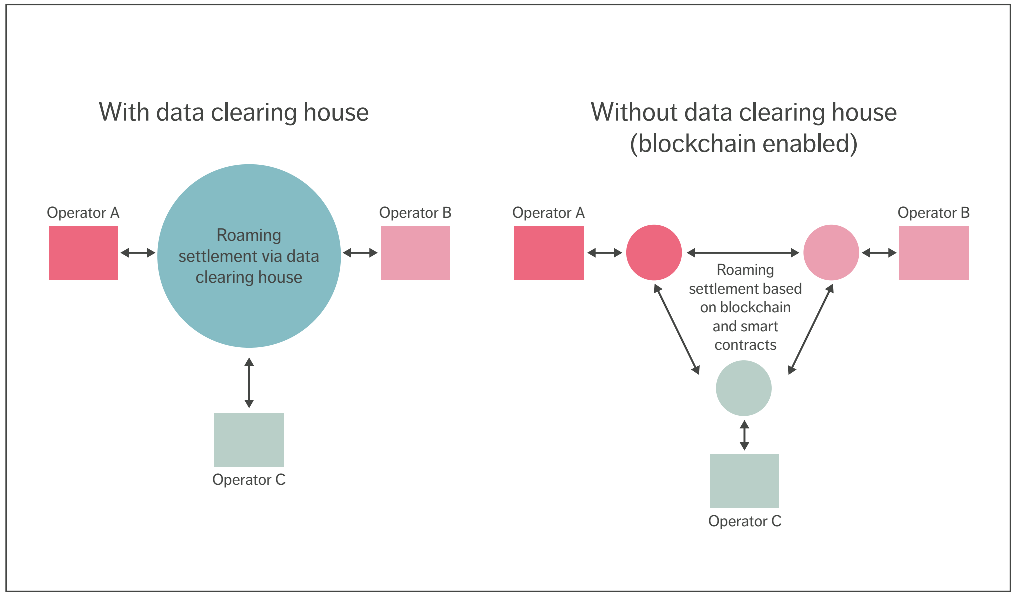

Figure 1 Roaming clearance and settlement, with and without third-party support

Smart contract platform for services providers

The smart contract platform is an innovation platform driven by Ericsson that allows operators who are innovating with us to explore blockchain and smart-contract technology to offer new services, evaluate platform business opportunities and address internal efficiencies to reduce the cost of doing business. One interesting use case for the platform is its application to roaming clearance and settlement services [1] as depicted in *Figure 1*.

The handling of roaming subscribers today relies on trusted third parties (data clearing companies, for example) to manage the clearing processes and settlement related to billing. The smart contract platform roaming settlement application replaces these (often expensive) third parties with a trusted, distributed and decentralized blockchain solution that includes smart contracts (for example, Hyperledger Fabric chain code).

The smart contract platform can take advantage of core attributes of blockchain's shared ledger approach to provide trust, security and transparency across the participating ecosystem. Smart contracts can be used to support the following three main groups of services:

- roaming management, including agreement definition and archiving
- data clearing, such as billing record creation, conversion services and fraud management
- financial clearing and settlement services for voice, SMS, MMS and data transactions.

The insights from smart contract platform experiments will validate the key technical properties where trust can be distributed and govern in a decentralized manner and through data integrity and transparency supported between the counterparties.

ID brokering

We have designed and implemented a decentralized system for ID brokering based on a concept that creates trust relations between digital identities and the systems that handle them. The system capitalizes on the strength of blockchains to express and manage trust relations in industry-wide solutions and creates a unified mechanism for ID management across underlying heterogeneous ID technologies.

ID brokering makes it easy to establish encrypted and trusted connectivity for IoT devices that are on the move, or for personal devices that are carried across different administrative network domains. For example, by allowing device IDs to act as digital passports and registering the (non-sensitive) passport IDs of devices when booking a trip, the networks the devices pa ss through (including airports, hotels and conference facilities) can use their own trusted IDs to grant secure internet access without manual authentication.

The ID brokering concept is based on three key aspects:

1. the self-sovereignty of ID domains, where devices are provisioned with any secure ID technology deemed appropriate, and where the ID secret is securely stored in a TEE
2. authentication utilizes the trust relation expressed in a blockchain-based backend, where instantaneous access rights for specific devices in specific networks are managed
3. the blockchain backend enables the system to reach a shared consensus on a global scale, as no single party is the main controller or beneficiary of the system.

Ericsson demonstrated an ID brokering implementation – in this case a custom layer on top of Hyperledger Fabric using blockchains and TEEs – at Mobile World Congress in 2017. In it, each IoT device is represented by a node, belongs to a domain, and has relations with owners expressed by links, as illustrated in *Figure 2*. With this approach, we emphasize the decentralized nature of applications enabled by the blockchain. Each domain owner

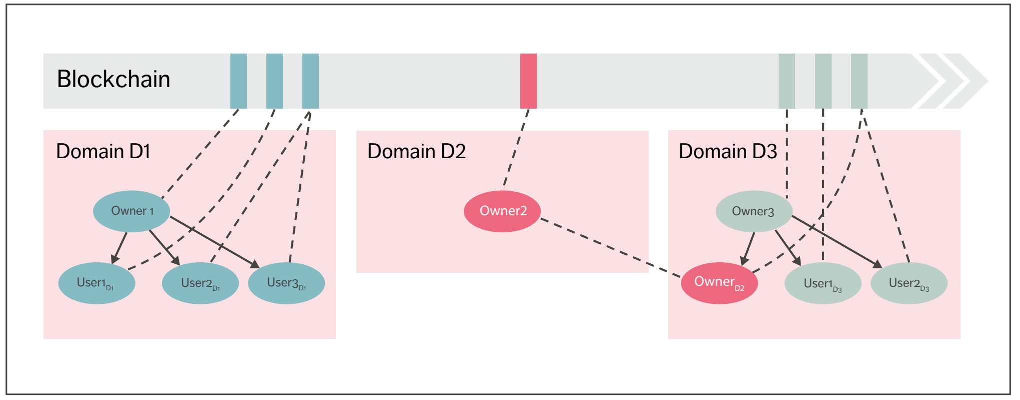

Figure 2 ID domain creation and ID crosslinking with the support of blockchain

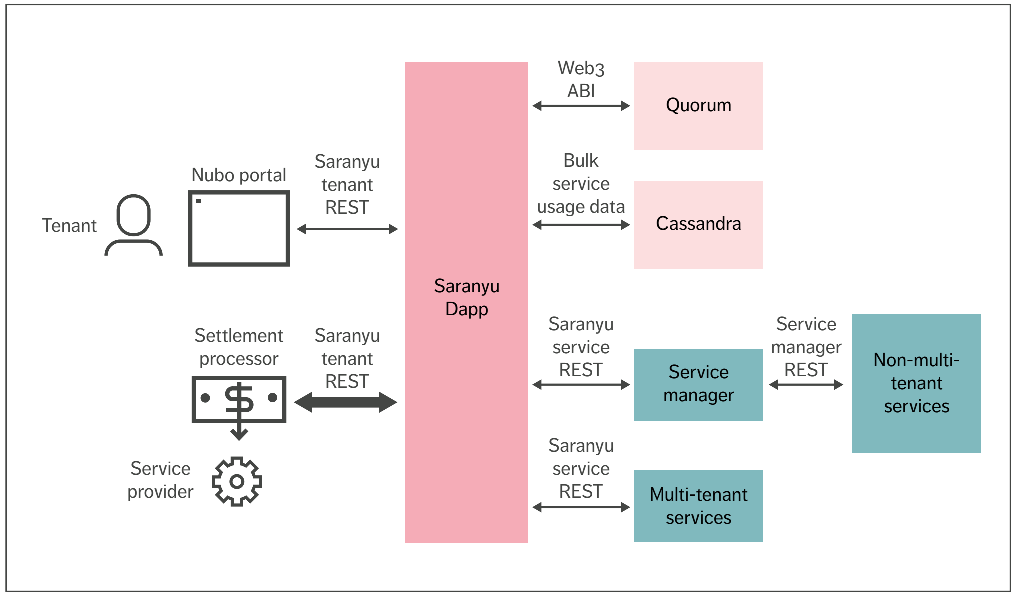

Figure 3 Nubo virtual services marketplace architecture

has full sovereignty of their domain, and shared context of the blockchain enables a domain to interact and to grant and revoke access dynamically.

The ID brokering solution shares the concept of self-sovereignty with the Sovrin system [2], and is oblivious to the specific ID technologies used for authentication and ID provisioning. Since 2017, we have been working on ID brokering and its coexistence with public key infrastructure solutions.

Nubo virtual services marketplace

New 5G features enable operator networks to be virtually segmented into different logical networks (slices) similarly to how network resources in cloud infrastructure can provide different virtual networks for different tenants. The rise of virtual network functions – that is, virtualized and software-based routers or firewalls – has created the foundation for a market of network services where the set of components can be composed specifically for each tenant. With slicing and virtualization of network components in 5G, we envision that future 5G operator services are likely to have similar characteristics, with a tailored composition of services for each network slice.

We designed the Nubo virtual services marketplace to meet the specific requirements of virtualization use cases. Its architecture is illustrated in *Figure 3*. The Nubo marketplace is made up of buyers of virtualized services, referred to as "tenants", and the sellers of those services, referred to as "service providers". The tenants can be individual users, enterprise customers or even operators. A blockchain with smart contracts provides the tenants with the basic trust platform for price discovery on the services. Nubo's tenant and service management microservice (known as Saranyu) utilizes the J.P. Morgan Quorum blockchain, which supports smart contracts written in Solidity.

Tenants and services have contract accounts on the blockchain, which govern their interaction with the marketplace and each other. Services list their resource offerings on the blockchain through Saranyu in the form of a JSON (JavaScript Object Notation) document describing the attributes of the resources. Attributes can be quota limited or have charges associated with them. Tenants request the delegation of resources and must cryptographically sign the JSON document, indicating that they commit to abide by the charging and quota advertised in the resource offerings.

Services record tenant usage and send usage records to Saranyu, which Saranyu stores in the Cassandra distributed database, depositing a signed hash of the record into the blockchain to ensure the records are not changed. Periodically, Saranyu runs a billing cycle in which tenant charges for services are totaled up and submitted to a settlement processor, which can be a credit card processor or a cryptocurrency account.

Nubo can also support cloud compute/networking/storage services as well as serverless functions or distributed operating system types of services. A prototype of Nubo was developed at Ericsson in 2018, featuring an experimental cryptocurrency charging system that charged for services using a private Ethereum account deployed in the Ericsson Research Data Center in Lund, Sweden. Services listed included the Nefele Cloud 3.0 distributed operating system, the Ethereum serverless function system, and the University of California, Berkeley, RISELab artificial intelligence execution environment Ray.

Standardization and external collaboration

The mass adoption of blockchains will require both technical and business-model interoperability between organizations, permissioned blockchain consortia, and even permissionless blockchains. Consequently, blockchain standardization is underway and several industry consortia have formed to strive for interoperability and harmonized processes. Ericsson is contributing to the standardization process through our active involvement in the GSMA and all major telecom and ICT standardization bodies, as well as by becoming a founding member in an ETSI (European Telecommunications Standards Institute) working group on permission distributed ledgers.

With respect to collaboration, the EU and several national governments are currently sponsoring academic and industrial collaboration for blockchain research and business acceleration. Ericsson has chosen to participate in the EU H2020-IoT SOFIE (Secure Open Federation for Internet Everywhere) project 2018-2020 together with several industry-leading companies and academic institutions to research blockchain interoperability across siloed IoT applications, including the demonstration of results through several live pilots. We are also collaborating directly with global technology companies in the areas of trusted computing and blockchains.

> THE MASS ADOPTION OF BLOCKCHAINS WILL REQUIRE BOTH TECHNICAL AND BUSINESS-MODEL INTEROPERABILITY

Conclusion

Ericsson sees significant value in blockchains as a trust enabler and potential disruptor that can enable completely new business models in the digital asset market. The use cases we have evaluated for private blockchains so far, both in-house and together with

> OUR NEXT STEPS WILL INCLUDE FURTHER EXPLORATION OF THE POTENTIAL OF PUBLIC BLOCKCHAINS AND HASHGRAPHS

global telco and enterprise customers, have achieved promising results. To date, we have demonstrated the value of blockchain for roaming settlement and other use cases such as IoT data monetization, supply chain management, handling of privacy-sensitive data, license management and ID management.

Our next steps will include further exploration of the potential of public blockchains and hashgraphs.

While we are keen to accelerate our blockchain efforts from exploration to commodification and mass adoption, we recognize that a number of fundamental issues must be resolved before we get there. Appropriate governance models around blockchain consortia must be established, for example, along with technology and business model interoperability. The questions of how to create a viable platform business and how to ensure that contracts act on trustworthy data must also be answered. We will continue to work on these aspects in close collaboration with our customers and other industry stakeholders through standardization and joint innovation.

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Service exposure:

A CRITICAL CAPABILITY IN A 5G WORLD

Exposure – and service exposure in particular – will be critical to the creation of the programmable networks that businesses need to communicate efficiently with Internet of Things (IoT) devices, handle edge loads and pursue the myriad of new commercial opportunities in the 5G world.

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While service exposure has played a notable role in previous generations of mobile technology – by enabling roaming, for example, and facilitating payment and information services over the SMS channel – its role in 5G will be much more prominent.

■ The high expectations on mobile networks continue to rise, with never-ending requests for higher bandwidth, lower latency, increased predictability and control of devices to serve a variety of applications and use cases. At the same time, we can see that industries such as health care and manufacturing have started demanding more customized connectivity to meet the needs of their services. While some of these demands can be met through improved network connectivity capabilities, there are other areas where those improvements alone will not be sufficient.

For example, in recent years, content delivery networks (CDNs) have been used in situations where deployments within the operator network became a necessity to address requirements like high bandwidth. More recently, however, new use-case categories in areas such as augmented reality (AR)/virtual reality (VR), automotive and Industry 4.0 have made it clear that computing resources need to be accessible at the edge of the network. This development represents a great opportunity for operators, enterprises and application developers to introduce and capitalize on new services. The opportunity also extends to web-scale providers (Amazon, Google, Microsoft, Alibaba and so on) that have invested in large-scale and distributed cloud infrastructure deployments on a global scale, thereby becoming the mass-market provider of cloud services.

Several web-scale providers have already started providing on-premises solutions (a combination of full-stack solutions and software-only solutions) to meet the requirements of certain use cases. However, the ability to expand the availability of web-scale services toward the edge of the operator infrastructure would make it possible to tackle a multitude of other use cases as well. Such a scenario is mutually beneficial because it allows the web-scale providers to extend the reach of services that benefit from being at the edge of the network (such as the IoT and CDNs), while enabling telecom operators to become part of the value chain of the cloud computing market.

> “SUCH A SCENARIO... [ENABLES] TELECOM OPERATORS TO BECOME PART OF THE VALUE CHAIN OF THE CLOUD COMPUTING MARKET”

Defining exposure

Exposure in the IT/telecom sphere can be divided into a number of subareas.

Data exposure is the process by which any kind of consumer (human or machine) can access data in a system via secure and controlled mechanisms. Data is normally exchanged in one direction only. Common examples of data exposure include accessing data via an application programming interface (API), downloading a file or retrieving observations from a server.

Service exposure goes beyond data exposure to also include the ordering of execution of operations in the underlying system. Using an API to initiate operations and/or processes is a good example of service exposure. Services can be invoked bidirectionally by triggering events, for example. Data can also be updated via a service.

Service exposure can be applied in a domain, as in **network exposure**, which exposes both data and services of the network. Enterprise resource planning (ERP) and customer relationship management (CRM) are other examples of domains where service exposure can be applied.

To maintain security, the details of the underlying system are typically hidden in exposure scenarios.

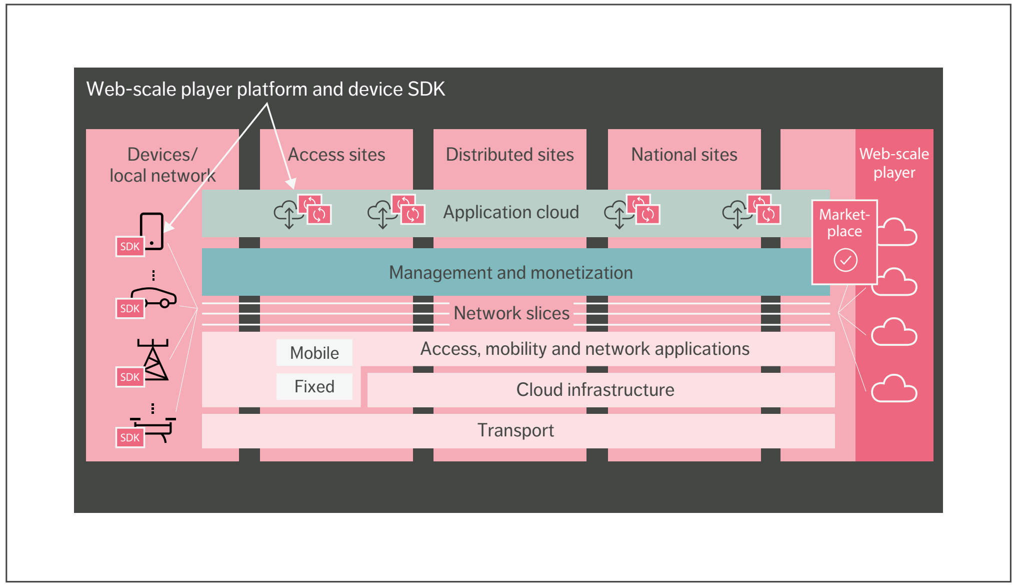

Figure 1 Collaboration with web-scale providers on telecom distributed clouds

Figure 1 illustrates how a collaboration with web-scale providers on telecom distributed clouds could be structured. We are currently exploring a partnership to enable system integrators and developers to deploy web-scale player application platforms seamlessly on telecom distributed clouds. Distributed cloud abstraction on the web-scale player marketplace encompasses edge compute, latency and bandwidth guarantee and mobility. Interworking with IoT software development kits (SDKs) and device management provides integration with provisioning certificate handling services and assignment to distributed cloud tenant breakout points.

In the mid to long term, service exposure will be critical to the success of solutions that rely on edge computing, network slicing and distributed cloud. Without it, the growing number of functions, nodes, configurations and individual offerings that those solutions entail represents a significant risk of increased operational expenditure. The key benefit of service exposure in this respect is that it makes it possible to use application programming interfaces (APIs) to connect automation flows and artificial intelligence (AI) processes across organizational, technology, business-to-business (B2B) and other borders, thereby avoiding costly manual handling. AI and analytics-based services are particularly good candidates for exposure and external monetization.

Key enablers

The 5G system architecture specified by 3GPP has been designed to support a wide range of use cases based on key requirements such as high bandwidth/throughput, massive numbers of connected devices and ultra-low latency. For example, enhanced mobile broadband (eMBB) will provide peak data rates above 10Gbps, while massive machine-type communications (mMTC) can support more than 1 million connections per square kilometer. Ultra-reliable low-latency communications (uRLLC) guarantees less than 1ms latency.

Fulfilling these eMBB, mMTC and uRLLC requirements necessitates significant changes to both the RAN and the core network. One of the most significant changes is that the core network functions (NFs) in the 5G Core (5GC) interact with each other using a Service-based Architecture (SBA). It is this change that enables the network programmability, thereby opening up new opportunities for growth and innovation beyond simply accelerating connectivity.

Service-based Architecture

The SBA of the 5GC network makes it possible for 5GC control plane NFs to expose Service-based Interfaces (SBIs) and act as service consumers or producers. The NFs register their services in the network repository function, and services can then be discovered by other NFs. This enables a flexible deployment, where every NF allows the other authorized NFs to access the services, which provides tremendous flexibility to consume and expose services and capabilities provided by 5GC for internal or external third parties. This support of the services subscription makes it completely different to the 4G/5G Evolved Packet Core network.

Because it is service-driven, SBA enables new service types and supports a wide variety of diversified service types associated with different technical requirements. 5G provides the SBI for different NFs (for example via SBI HTTP/2 Restful APIs). The SBI can be used to address the diverse service types and highly demanding performance requirements in an efficient way. It is an enabler for short time to market and cloud-native web-scale technologies.

The 3GPP is now working on conceptualizing 5G use cases toward industry verticals. Many use cases can be created on-demand as a result of the SBA.

Distributed cloud infrastructure

The ability to deploy network slices – an important aspect of 5G – in an automated and on-demand manner requires a distributed cloud infrastructure. Further, the ability to run workloads at the edge of the network requires the distributed cloud infrastructure to be available at the edge. What this essentially means is that distributed cloud deployments within the operator network will be an inherent part of the introduction of 5G. The scale, growth rate, distribution and network depth (how far out in the network edge) of those deployments will vary depending on the telco network in question and the first use cases to be introduced.

As cloud becomes a natural asset of the operator infrastructure with which to host NFs and services (such as network slicing), the ability to allow third parties to access computing resources in this same infrastructure is an obvious next step. Contrary to the traditional cloud deployments of the web-scale players, however, computing resources within the operator network will be scarcer and much more geographically distributed. As a result, resources will need to be used much more efficiently, and mechanisms will be needed to hide the complexity of the geographical distribution of resources.

> CORE NETWORK FUNCTIONS IN THE 5GC INTERACT WITH EACH OTHER USING A SERVICE-BASED ARCHITECTURE

Cloud-native principles

The adoption of cloud-native implementation principles is necessary to achieve the automation, optimized resource utilization and fast, low-cost introduction of new services that are the key features of a dynamic and constrained ecosystem. Cloud-native implementation principles dictate that software must be broken down into smaller, more manageable pieces as loosely coupled stateless services and stateful backing services. This is usually achieved by using a microservice architecture, where each piece can be individually deployed, scaled and upgraded. In addition, microservices communicate through well-defined and version-controlled network-based interfaces, which simplifies integration with exposure.

Three types of service exposure

There are three main types of service exposure in a telecom environment:

- network monitoring
- network control and configuration
- payload interfaces.

Examples of network monitoring service exposure include network publishing information as real-time statuses, event streams, reports, statistics, analytic insights and so on. This also includes read requests to the network.

Service exposure for network control and configuration involves requesting control services that directly interact with the network traffic or request configuration changes. Configuration can also include the upload of complete virtual network functions (VNFs) and applications.

Examples of service-exposure-enabled payload interfaces include messaging and local breakout, but it should be noted that many connectivity/payload interfaces bypass service exposure for legacy reasons. Even though IP connectivity to devices is a service that is exposed to the consumer, for example, it is currently not achieved via service exposure. The main benefit of adding service exposure would be to make it possible to interact with the data streams through local breakout for optimization functions.

Leveraging software development kits

At Ericsson, we are positioning service exposure capabilities in relation to developer workflows and practices. Developers are the ones who use APIs to create solutions, and we know they rely heavily on SDKs. There are currently advanced developer frameworks for all sorts of advanced applications including drones, AR/VR, the IoT, robotics and gaming. Beyond the intrinsic value in exposing native APIs, an SDK approach also creates additional value in terms of enabling the use of software libraries, integrated development environments (IDEs) plug-ins, third-party provider (3PP) cloud platform extensions and 3PP runtimes on edge sites, as well as cloud marketplaces to expose these capabilities.

Software libraries can be created by prepackaging higher-level services such as low-latency video streaming and reverse charging. This can be achieved, for example, by using the capabilities of network exposure functions (NEF) and service capability exposure functions (SCEF), creating ready-to-deploy functions or containers that can be distributed through open repositories, or even marketplaces, in some cases. This possibility is highly relevant for edge computing frameworks.

Support for IDE plug-ins eases the introduction of 3PP services with just a few additional clicks. Selected capabilities within 3PP cloud platform extensions can also create value by extending IoT device life-cycle management (LCM) for cellular connected devices, for example. The automated provisioning of popular 3PP edge runtimes on telco infrastructure enables 3PP runtimes on edge sites.

> CLOUD MARKETPLACES ARE AN IDEAL PLACE TO EXPOSE ALL OF THESE CAPABILITIES

Finally, cloud marketplaces are an ideal place to expose all of these capabilities. The developer subscribes to certain services through their existing account, gaining the ability to activate a variety of libraries, functions and containers, along with access to plug-ins they can work with and/or the automated provisioning required for execution.

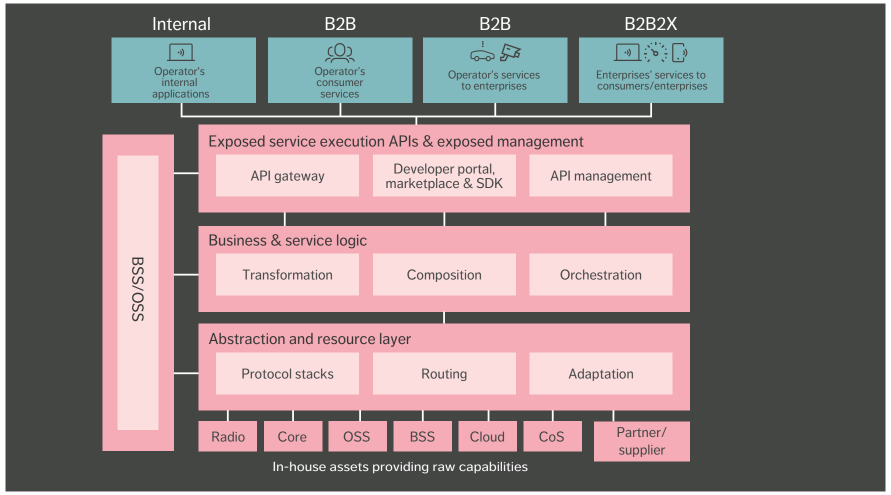

Figure 2 Functional architecture for service exposure

Functional architecture for service exposure

The functional architecture for service exposure is built around four customer scenarios:

- internal consumers
- business-to-consumers (B2C)
- business-to-business (B2B)
- business-to-business-to-business/consumers (B2B2X).

In the case of internal consumers, applications for monitoring, optimization and internal information sharing operate under the control and ownership of the enterprise itself. In the case of B2C, consumers directly use services via web or app support. B2C examples include call control and self-service management of preferences and subscriptions. The B2B scenario consists of partners that use services such as messaging and IoT communication to support their business. The B2B2X scenario is made up of more complex value chains such as mobile virtual network operators, web scale, gaming, automotive and telco cloud through web-scale APIs.

Figure 2 illustrates the functional architecture for service exposure. It is divided into three layers that each act as a framework for the realization. Domain-specific functionality and knowledge are applied and added to the framework as configurations, scripts, plug-ins, models and so on. For example, the access control framework delivers the building blocks for specializing the access controls for a specific area.

The abstraction and resource layer is responsible for communicating with the assets. If some assets are located outside the enterprise – at a supplier or partner facility in a federation scenario, for example – B2B functionality will also be included in this layer.

The business and service logic layer is responsible for transformation and composition – that is, when

SERVICE EXPOSURE IN 5G ✱

> COMMON EXPOSURE FUNCTIONS [CAN BE DEPLOYED] BOTH IN A DISTRIBUTED WAY AND INDIVIDUALLY

there is a need to raise the abstraction level of a service to create combined services.

The exposed service execution APIs and exposed management layer are responsible for making the service discoverable and reachable for the consumer. This is done through the API gateway, with the support of portal, SDK and API management.

Business support systems (BSS) and operations support systems (OSS) play a double role in this architecture. Firstly, they serve as resources that can expose their values – OSS can provide analytics insights, for example, and BSS can provide "charging on behalf of" functionality. At the same time, OSS are responsible for managing service exposure in all assurance, configuration, accounting, performance, security and LCM aspects, such as the discovery, ordering and charging of a service.

One of the key characteristics of the architecture presented in Figure 2 is that the service exposure framework life cycle is decoupled from the exposed services, which makes it possible to support both short- and long-tail exposed services. This is realized through the inclusion and exposure of new services through configuration, plug-ins and the possibility to extend the framework.

Another key characteristic to note is that it is possible to deploy common exposure functions both in a distributed way and individually – in combination with other microservices for efficiency reasons, for example. Typical cases are distributed cloud with edge computing and web-scale scenarios such as download/upload/streaming where the edge site and terminal are involved in the optimization.

The exposure framework is realized as a set of loosely connected components, all of which are cloud-native compliant and microservice based, running in containers. There is not a one-size-fits-all deployment – some of the components are available in several variants to fit different scenarios. For example, components in the API gateway support B2B scenarios with full charging but there are also scaled-down versions that only support reporting, intended for deployment in internal exposure scenarios.

Other key properties of the service exposure framework are:

Terms and abbreviations

3PP – Third-party Provider | **5GC** – 5G Core | **AI** – Artificial Intelligence | **API** – Application Programming Interface | **AR** – Augmented Reality | **B2B** – Business-to-Business | **B2BCX** – Business-to-Business-to-Business/Consumers | **B2C** – Business-to-Consumers | **BSS** – Business Support Systems | **CDN** – Content Delivery Network | **CoS** – Communication Services | **CRM** – Customer Relationship Management | **eMBB** – Enhanced Mobile Broadband | **ERP** – Enterprise Resource Planning | **IDE** – Integrated Development Environment | **IoT** – Internet of Things | **LCM** – Life-cycle Management | **mMTC** – Massive Machine-type Communications | **NEF** – Network Exposure Functions | **NF** – Network Function | **ONAP** – Open Network Automation Platform | **OSS** – Operations Support Systems | **SBA** – Service-based Architecture | **SBI** – Service-based Interface | **SCEF** – Service Capability Exposure Functions | **SDK** – Software Development Kit | **uRLLC** – Ultra-reliable Low-latency Communications | **VNF** – Virtual Network Function | **VR** – Virtual Reality

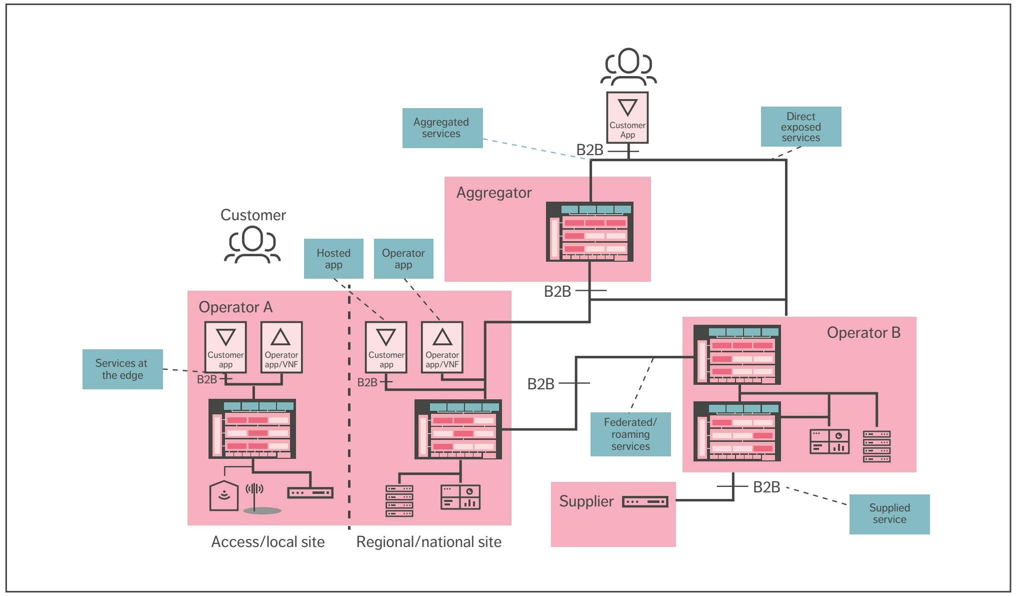

Figure 3 Service exposure deployment (dark pink boxes indicate deployed components)

- scalability (configurable latency and scalable throughput) to support different deployments
- diversified API types for payload/connectivity, including messaging APIs (request-response and/or subscribe-notify type), synchronous, asynchronous, streaming, batch, upload/download and so on
- multiple interface bindings such as restful, streaming and legacy
- multivendor and partner support (supplier/federation/aggregator/web-scale value chains)
- security and access control functionality.

Deployment examples

Service exposure can be deployed in a multitude of locations, each with a different set of requirements that drive modularity and configurability needs. *Figure 3* illustrates a few examples.

In the case of Operator B in Figure 3, service exposure is deployed to expose services in a full B2B context. BSS integration and support is required to handle all commercial aspects of the exposure and LCM of customers, contracts, orders, services and so on, along with charging and billing. Operator B also uses the deployed B2B commercial support to acquire services from a supplier.

In the case of Operator A, service exposure is deployed both at the central site and at the edge site to meet latency or payload requirements. Services are only exposed to Operator A's own applications/VNFs, which limits the need for B2B support. However, due to the fact that Operator A hosts some applications for an external partner, both centrally and at the edge, full B2B support must be deployed for the externally owned apps.

The aggregator in Figure 3 deploys the service exposure required to create services put together by more than one supplier. Unified Delivery Network and web-scale integration both fall into this category. As exposure to the consumer is done through the aggregator, this also serves as a B2B interface to handle specific requirements. Examples of this include the advertising and discovery of services via the portals of web-scale providers.

A subset of B2B support is also deployed to provide the service exposure that handles the federation relationship between Operator A and Operator B, in which both parties are on the same level in the ecosystem value chain.

Conclusion

There are several compelling reasons for telecom operators to extend and modernize their service exposure solutions as part of the rollout of 5G. One of the key ones is the desire to meet the rapidly developing requirements of use cases in areas such as the Internet of Things, AR/VR, Industry 4.0 and the automotive sector, which will depend on operators' ability to provide computing resources across the whole telco domain, all the way to the edge of the mobile network. Service exposure is a key component of the solution to enable these use cases.

Recent advances in the service exposure area have resulted from the architectural changes introduced in the move toward 5G and the adoption of cloud-native principles, as well as the combination of Service-based Architecture, microservices and container technologies. As operators begin to use 5G technology to automate their networks and support systems, service exposure provides them with the additional benefit of being able to use automation in combination with AI to attract partners that are exploring new, 5G-enabled business models. Web-scale providers are also showing interest in understanding how they can offer their customers an easy extension toward the network edge.

Modernized service exposure solutions are designed to enable the communication and control of devices, providing access to processes, data, networks and OSS/BSS assets in a secure, predictable and reliable manner. They can do this both internally within an operator organization and externally to a third party, according to the terms of a Service Level Agreement and/or a model for financial settlement.

Service exposure is an exciting and rapidly evolving area and Ericsson is playing an active role in its ongoing development. As a complement to our standardization efforts within the 3GPP and Industry 4.0 forums, we are also engaged in open-source communities such as ONAP (the Open Network Automation Platform). This work is important because we know that modernized service exposure solutions will be at heart of efficient, innovative and successful operator networks.

Further reading

- **Ericsson web page, Service enablement, available at:**
  https://www.ericsson.com/en/portfolio/digital-services/cloud-core/service--enablement
- **Ericsson web page, Cloud core exposure server, available at:**
  https://www.ericsson.com/en/portfolio/digital-services/cloud-core/cloud-unified-data-management-and-policy/cloud-core-exposure-server
- **Ericsson web page, Cloud packet core, available at:**
  https://www.ericsson.com/en/portfolio/digital-services/cloud-core/cloud-packet-core

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SIX KEY TRENDS MANIFESTING THE PLATFORM FOR INNOVATION

TECHNOLOGY TRENDS 2019

BY: ERIK EKUDDEN, CTO

Affordable and efficient connectivity is a fundamental component of digitalization and has become as important as clean water and electricity in creating a sustainable society of the future. Recognition of this fact is of critical importance as we enter a new era that is defined by the combinatorial effects of a multitude of transformative technologies in areas such as mobility, the Internet of Things (IoT), distributed computing and artificial intelligence (AI).

The universal connectivity network that we use today is built on voice and mobile broadband services that currently serve 9 billion connected devices globally. This technology is recognized and acknowledged for its availability, reliability, integrity and affordability, and it is trusted to handle sensitive and important information. Today's network provides pervasive global coverage on a scale with which no other technology can compete. It has quickly become a multipurpose network, ready and able to onboard all types of users, as well as supporting a large number of new use cases and a plethora of new technologies to meet any consumer or enterprise need. As such, it is ideally suited to serve as the foundation for future innovation in any application.

APPROPRIATE AND UNIVERSAL CONNECTIVITY

The multipurpose network is significantly more cost-efficient than specialized or dedicated network solutions, making it the most affordable solution to address society's needs across the spectrum from human-to-human to human-to-thing and thing-to-thing communication. It supports everything from traditional voice calls to immersive human-to-human communication experiences. In terms of human-to-thing communication, it enables everything from digital payments to voice-controlled digital assistants, as well as real-time sensitive drone control and high-quality media streaming.

With regard to IoT communication, the ubiquitous connectivity provided by the multipurpose network enables the creation of a physical world that is fully automated and programmable. Examples of this include massive sensor monitoring, fully autonomous physical processes such as self-driving cars and manufacturing robots, as well as digitally-embedded processes such as autonomous decision-making in tax returns.

KEY TECHNOLOGY TRENDS

In my view, the ongoing evolution toward the future network continues to rely heavily on the five key technology trends that I outlined in last year's trends article. Therefore, in this year's technology trends article, I have chosen to build on last year's conclusions and share my view of the future network platform in relation to those five trends, with one addition: distributed compute and storage.

Demanding use cases exemplified by trends 1 and 2

Today's networks are transforming into a platform where applications, processes and other technologies are developed, deployed and enhanced. For me, it is fundamental that the platform ensures affordable, reliable and trusted operation. Two use cases that I expect the network platform will need to support are trends 1 and 2: the Internet of Skills and cyber-physical systems (CPSs).

TREND #1: INTERNET OF SKILLS

The Internet of Skills has the potential to bridge the geographical distance between humans as well as between humans and things. A high quality of experience (QoE) is essential to create immersive interactions that allow humans to attend meetings remotely with the same ability to participate as if they were physically present. Humans have to trust the network to enable critical remote operations and interaction with things.

Self-driving vehicles will require a remote person to take over the driving or support in the decision-making if the autonomous system fails. Hence, tele-operation of robots and vehicles is needed at sea, on land and underground, as well as in the air. Remote human assistance is also required for tasks such as maintenance, troubleshooting and repairing across industrial, enterprise, health care and consumer domains. The Internet of Skills also applies to the ability to experience physical items remotely in applications such as online shopping and gaming.

High-quality and efficient capturing, transmission and rendering of visual, audio and haptic information is essential to the Internet of Skills. This information will be captured by multiple devices and it must be fused together to be reproduced remotely. A distributed environment for access, compute and storage of this information is therefore highly advantageous.

Haptic communications require latencies below 10ms in the most demanding scenarios. Large volumes of 3D visual data and high-frequency haptic data impose high network bandwidth and latency demands, both in the uplink and downlink.

A network platform with low-latency characteristics allows for large amounts of data to be quickly transmitted between devices. This means that more time can be spent on processing and performing analytics on the available information to enhance the experience.

Security and privacy are very important since the devices may capture sensitive visual, audio and haptic information. This information can relate to the user of the device or other users that share the same environment, including detailed characteristics of the user's physical environment such as their home or office, as well as insights into the user's daily activities.

The network platform will also be very beneficial for enabling the positioning of devices, both outdoors and indoors. The network radio positioning information can be fused with information from the device's onboard sensors such as the camera and inertial sensors.

TREND #2: CYBER-PHYSICAL SYSTEMS

CPS results from the integration of different systems to control a physical process and uses feedback to adapt to new conditions in real time. This is achieved by integrating physical processes, networking and computation. A CPS generates and acquires data, so that the relevant elements involved have access to the appropriate information at the right time. Therefore, the CPS can autonomously determine its current operating status, and corrective actions are realized by the actuators. Information comes from sensors and from other related CPSs. The role of humans is to supervise the operation of the automated and self-organizing processes.

Communication is vital in CPSs to allow different and heterogeneous objects to exchange information with each other and with humans, at any time and in any conditions. Deterministic communication (in terms of latency, bandwidth and reliability) largely impacts the dynamic interactions between subsystems in CPSs. Minimizing the time it takes to perform control tasks is critical to ensuring that a system functions correctly.

The future network platform should provide the specific connectivity performance to guarantee CPS-critical requirements. As an example, latency criticality is an issue for all cases where a controller or complex AI must take decisions and actions in real time.

Each CPS has a specific architecture that requires an adaptive network platform. Hence, a specific ad-hoc design of indoor and/or outdoor coverage is required. In addition, network slicing will enable satisfying heterogeneous connectivity requirements on the same network, for any indoor or outdoor scenarios.

EXAMPLES OF CYBER-PHYSICAL SYSTEMS

PORTS OF THE FUTURE

Terminal port operations will increasingly consist of a mixture of physical machinery, robotics systems, automated vehicles, human-operated digital platforms and AI-based software systems. These elements will transform future ports into CPSs, creating a digital ecosystem comprised of various intelligent agents highly specialized in specific aspects of cargo loading/unloading and of the logistic chains.

AUTOMOTIVE

All new features in modern cars, such as advanced driver assistance systems and connected vehicle services, are based on electronics and software rather than on mechanical engineering innovations. Safety-critical functions, driver-assistance software and infotainment applications will run in specific and highly compartmentalized onboard modules that interact with a plethora of sensors and actuators. In this context, the future vehicle will increasingly take the form of a CPS for which the prevention of accidents is the main goal.

SMART MANUFACTURING

The factory of the future will be a set of interacting CPSs, where highly skilled workers will have direct insight into the operations of coordinated intelligent machines from a central control entity. Every functional aspect of a production chain will be affected – from design, to manufacturing, through to supply chains, and later extending to customer service and support. The smart factory will be hyper-connected, data-intensive and highly secure.

My vision of the future network platform

As I see it, the future network platform is characterized by its capability to instantaneously meet any application needs. It can handle huge amounts of data, scarce amounts of data, and everything in between. It will meet requirements for both open data and sensitive data, as well as all manner of needs related to uplink and downlink transmission. From real-time critical to non-critical, predefined to flexible air interface, preset to adaptive routing – the future network platform has it covered. Anyone and anything that can benefit from a connection should be able to access and use the network.

MAIN CHARACTERISTICS

The interconnect between different kinds of networks, from local to wide-area coverage, builds a global network that provides a platform for pervasive global services. The inherent mobility within and between the networks creates unprecedented coverage both indoors and outdoors. Utilizing all these network assets enables a distributed environment for access, compute and storage. These assets are virtualized, distributed across the network, and are made available where they are needed and are most efficient. Applications and processes are dynamically deployed throughout the network. Network slicing enables streamlined connections for different applications, enhancing the efficiency of the total usage of the network.

Autonomous deployment, operation and orchestration is an essential capability of the network platform to enable cost-efficiency. Just as important are the reliability and resilience to fulfill expectations from industry and society. Built-in, automated security functions protect the network and the integrity of its users from external threats.

THE NETWORK PLATFORM OFFERING

The network platform offers a wide range of capabilities to all its users. It provides a seamless universal connectivity fabric with almost unlimited, scalable and affordable distributed compute and storage. Sensors and actuators can be attached anywhere throughout the network. Latency can be optimized by interacting with the control of access, compute and storage. Embedded into the platform is a distributed intelligence that supports users with insights and reasoning.

The addressability and reachability capabilities make it possible to connect anyone or anything regardless of location and time. Together with the inherent security and availability, the network platform can also meet communication needs relating to secure identification of users and networks. It also provides the scalability to automatically adapt to the exact needs of individual users and applications. As an example, adaptive power consumption is enabled by a flexible air interface. Another example is automated life-cycle management of devices, users and applications. This guarantees the most cost-efficient solution for users, in both the long and short term.

The network platform offering is consumed through an automated digital marketplace. Network services and data are available through consistent and open business interfaces for the applications (APIs). Data, such as location, connectivity conditions and user behavior, can be made available from the network platform.

With all these capabilities, the network platform offers the most accessible and valuable foundation for future innovation.

Four technologies evolving the network platform: Trends 3-6

In my view, four technology areas are crucial to the evolution of the future network platform, represented by trends 3 to 6: distributed compute and storage, ubiquitous radio access, security assurance and zero-touch networks.

TREND #3: DISTRIBUTED COMPUTE AND STORAGE

Future applications will require new processing capabilities from the network in order to reduce the amount of data that needs to be communicated, provide low latency, and increase robustness and security.

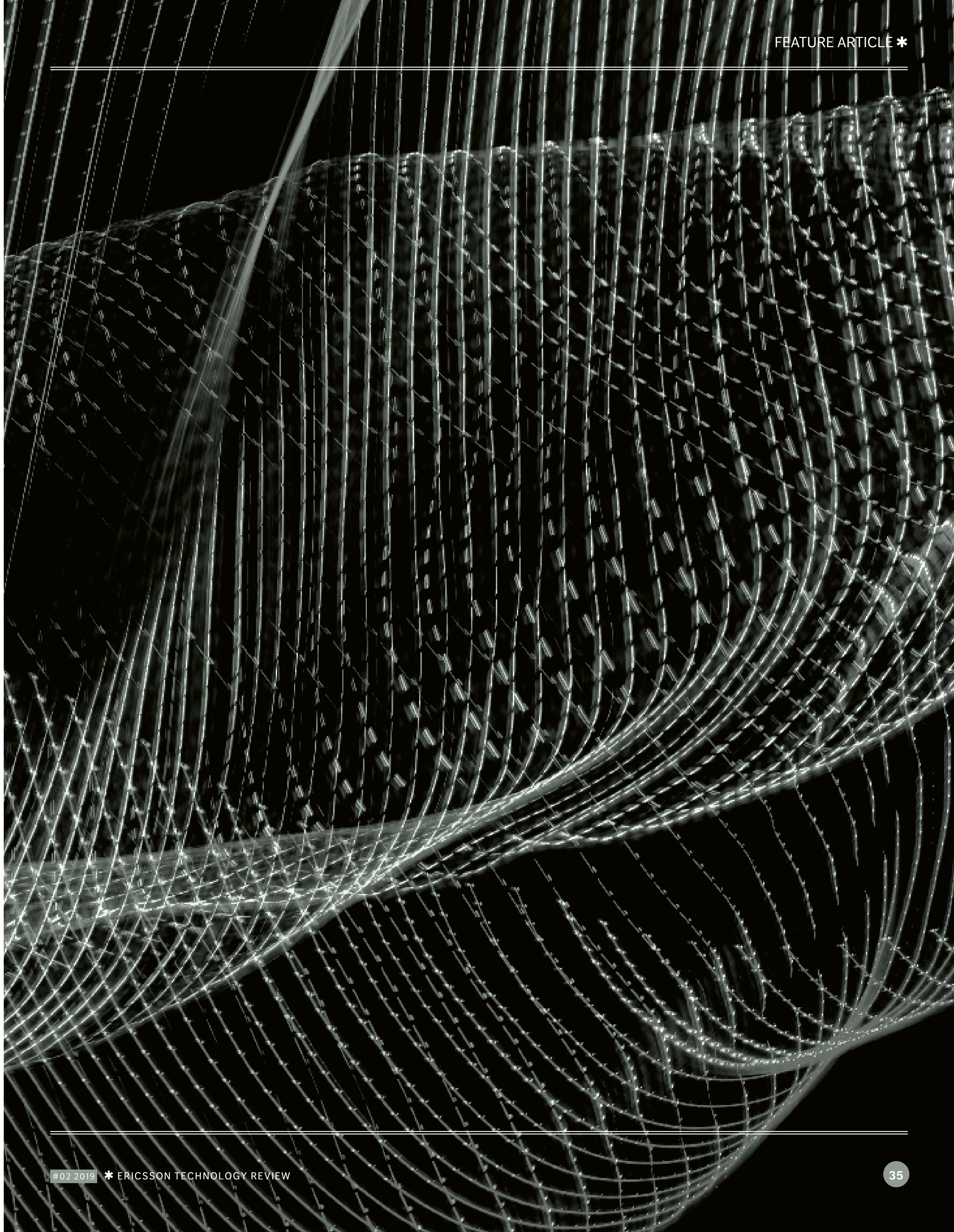
Today's processors and accelerators will eventually experience the end of Moore's Law, and new heterogeneous computing solutions will emerge. Commodity hardware has been joined by a highly heterogeneous set of specialized chipsets – often referred to as accelerators – that are optimized for a certain class of applications. For example, data-intensive applications such as machine learning (ML)/AI or augmented reality/virtual reality can take advantage of the massive parallelization offered by Graphical Processing Units or Tensor Processing Units. Latency-sensitive applications can utilize computation pattern reuse offered by either custom-designed integrated circuits or field-programmable integrated circuits.

The next step of heterogeneous computing will involve new computing paradigms such as neuromorphic processors that yield low power consumption, fast inference and event-driven information processing. Another emerging technology is photonic computing. Photons are used instead of electrons, thus avoiding the latency of the electron-switching times. Quantum processor-based acceleration of compute-intensive and latency-sensitive algorithms will eventually become a reality. By exploiting the quantum mechanics principles such as superposition and entanglement, quantum processors promise exponential growth of computing power for a certain class of problems.

The emergence of universal memories will offer the capacity and persistency features of storage, combined with byte-addressability and increased access speed of memory. Programs written for persistent memories can remove the distinction between runtime data structures and offline data storage structures, resulting in faster start-up times and recovery in case of failover. Advancements in non-volatile memory technologies will be crucial to meet strict latency requirements.

The increasing disparity of central processing unit speeds versus memory access speeds will lead to memory-centric compute architectures. Compute units will be embedded inside the memory or the storage fabrics. This will not only increase performance, but also lead to significant energy-efficiency gains by reducing the data movement of traditional compute-centric architectures.

Efficiently developing applications for a distributed compute environment will require new programming models. Programs will benefit from separating the intent of the application from the how



and where of the physical network. Today, intent-based networking uses Service Level Agreements and policies to define the intent of network operations. The network configures, monitors and troubleshoots issues in the network to fulfill these intents. In the future, there will be more cloud services managed by intent-based operations to evolve toward more advanced automation.

The network platform will benefit from the seamless integration of specialized compute and storage hardware to boost performance for a wider range of emerging, complex applications. The advanced compute and storage capabilities will be moved to the edge of the network, closer to where the data is generated. Further, the network will be able to support developers with efficient and transparent programming models. Edge-native applications will be designed from the ground up to fully capitalize on compute and storage resources anywhere.

TREND #4: UBIQUITOUS RADIO ACCESS

Improved indoor coverage, maximal energy efficiency, fiber-like performance and support for both small cells and a wide range of new use cases are key features of the 5G networks that are currently being rolled out. These networks will be the baseline for future radio networks and the network platform itself.

Future wireless access networks will consist of a wide range of different types of nodes jointly providing wireless access coverage. Devices will in many cases have simultaneous connectivity to multiple network nodes, including different access technologies, for enhanced performance and reliability. Wireless technology will also be used for the connectivity between the network nodes, as a complement to fiber-based connectivity.

Network coverage will be further extended by making use of intermediate devices to forward data to devices outside the coverage of the basic network. Device cooperation can be used to create virtual large antenna array by combining the antennas of multiple devices, which requires tight synchronization. As the network is becoming increasingly dense with a greater amount of small low-power network nodes, and with devices contributing to the overall connectivity, the border between devices and network nodes may be more diffuse.

Key to the management of this kind of massive heterogenous network, with a much more mesh-like connectivity, will be the development and utilization of advanced AI functionality. This will enable the network to evolve and adapt over time to new requirements and changes in the environment.

Operation above 100GHz will enable terabit-per-second data rates, although only for truly short-range connectivity. There are currently implementation challenges for this frequency range, such as how to generate substantial power and the heat dissipation, considering the inherently small dimensions of the components, including antennas. The extension to higher-frequency operation and use of beam-formed transmissions will enable enhancements in spectrum sharing.

In the higher layers of RANs and core networks, the evolution toward cloud-native implementation and automation continues. Network interfaces are moving away from traditional point-to-point interfaces toward more services-based application interfaces decoupled from underlying transport connections. Cloud-native implementation of stateless network functions use external context storage for redundancy and context management for different events, such as context relocation when mobile.

Beyond the primary task of providing wireless connectivity, the radio-access infrastructure will also be capable of delivering other services. This is already happening today, in part, with the introduction of location-based services as a complement to GPS. The combination of high-frequency band networks and dense deployments will make it possible to dramatically enhance the accuracy down to sub-meter level. Other service examples include time synchronization, time-sensitive networking, the collection of complementary information about local weather conditions and the creation of radar-like scans of the environment.

TREND #5: SECURITY ASSURANCE

The need for protection and assurance (or even compliance) is growing rapidly as business and society increasingly rely on universal connectivity and compute. Today, there is intense activity to explore the potential of AI and ML to protect systems and networks. There is large-scale adoption of these technologies in areas such as network threat detection and threat intelligence extraction, while other areas such as continuous authentication appear less mature. While AI technologies can provide a wide range of benefits, it is important to note that they can also be used by adversaries to find avenues of attack that specifically target ML systems.

In these autonomous networks, security assurance procedures play the important role of verifying security properties of the network platform. One challenge lies in the network architectures, based on cloud technologies and virtualization, which are introducing requirements for continuous compliance verification in a dynamic environment. At the same time, security assurance needs to be rooted in the evidence collected in the network slices supporting different industries. AI and ML technologies will bring automation of assurance and compliance verification to the network platform.

In the world of cloud computing, enclave and confidential computing hardware solutions that provide a root of trust are currently being packaged in pre-commercial cloud solutions. These technologies have the potential to become prevalent when addressing security concerns for processing in the cloud. Conceptually similar trusted computing technologies are also moving into IoT devices.

The trend toward encryption everywhere continues with reports of up to 90 percent usage of HTTPS. A substantially different protocol stack on the internet is expected in a few years, with QUIC and DoH as the dominant protocols, protected by newly standardized post-quantum algorithms. Since current security protocols are not suited for constrained IoT nodes and devices, the industry is working to standardize new lightweight application layer protocols.

At the same time, remotely managed eUICC (embedded Universal Integrated Circuit Card) based SIM identities in IoT devices are increasingly being deployed for network access. Modern SIMs based on the eUICC, and later the even more cost-effective iUICC (integrated Universal Integrated Circuit Card), will form the trust anchors for secure identities and network access in five to seven years.

Mission-critical use cases and regulatory demands, as well as cloud and edge computing, are the driving forces behind the trust and assurance technologies that are being developed and becoming integral parts of the network platform.

TREND #6: ZERO-TOUCH NETWORKS

A zero-touch network is capable of self-management and is controlled by business intents. Data-driven control logic makes it possible to design the system without the need for human configuration, as well as to provide a higher degree of information granularity. Applying AI technologies will enable zero-touch automation of network life-cycle management, including optimizing system performance, predicting upcoming faults and enabling preventive actions.

The performance of a data-driven zero-touch function can increase by utilizing the wider network data from many local clients, but this needs to be balanced against the cost and time associated with transferring large volumes of data.

One approach is to design distributed ML solutions, such as federated learning, which makes it possible to generate a network-wide global ML model. Training is done on local clients, and the need to transfer data is limited to model updates, instead of raw data.

With reinforcement learning, it is possible to design a solution that responds to unforeseen environments, which can be used to automate or optimize a specific process. A reinforcement learning agent learns how to act optimally given the system state information and reward function, focusing on finding a balance between exploration of uncharted territory and exploitation of current knowledge. The requirements on reliability and safety will, however, set limits on the applicability.

Robots are used to interface with the network infrastructure, collaborate with humans and utilize AI to perform physical inspections, determine fault causes, predict future faults and plan maintenance work. Computer-vision techniques enable, for example, automated cell tower inspection, while machine reasoning is used to plan and execute drone flight. Techniques to generalize and transfer lessons learned can be used to increase performance from one tower inspection to another. These AI-based robot systems will collaborate with humans, thereby increasing their safety and efficiency.

An intent-based approach similar to the one referenced in trend 3 (distributed compute and storage) allows human users to interact with the AI system that is part of zero-touch applications. Domain modeling, knowledge representation and reasoning (together with ML) are used to create a cognitive layer for humans to interact with the system using high-level intents. The system is capable of evaluating and executing strategies in line with an intent, based on lower-level key performance indicator (KPI) predictions. By complementing ML with machine reasoning, the system can be designed to express why certain decisions were taken and is a way to implement explainable AI.

Trustworthy ML models that fulfill zero-touch aspects need to be built in line with the need for privacy and legislative rules for how data can be exposed or moved. New specialized hardware for accelerating ML training and inference will improve performance and reduce energy consumption in a well-designed zero-touch network platform. Recent progress in AI has shown new promising possibilities to design for zero touch. Many challenges need to be overcome, however, and the value and efficiency of traditionally designed control logic should not be underestimated.

CONCLUSION

No other technology in the world today can provide pervasive global coverage on a scale comparable to that of the network platform, and it is my firm belief that it is ideally suited to serve as the innovation platform for both current and future applications. The technology evolution characterized by this year's trends points toward the future definition of 6G.

Much more cost-efficient than specialized or dedicated network solutions, the network platform is clearly the most affordable solution to address society's needs across the spectrum from human-to-human to human-to-thing and thing-to-thing communication. One of its major advantages is that it is available through an open marketplace that is accessible to anyone, anywhere, at any time.

The multipurpose network is rapidly emerging as a secure, robust and reliable platform where applications, processes and other technologies can be developed, deployed and managed. The Internet of Skills and cyber-physical systems – trends 1 and 2 – are important examples of use cases that it needs to support.

A key characteristic of the future network platform will be its ability to instantaneously meet any application need, anytime. Four technology areas – trends 3-6 – are playing critical roles in its ongoing evolution: distributed compute and storage, ubiquitous radio access, security assurance and zero-touch networks.

Self-driving vehicles, intelligent manufacturing robots and real-time drone control are just a few examples of the myriad of ways in which the multipurpose network is enabling the automation of the physical world and, ultimately, the creation of a sustainable society of the future.

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ERIK EKUDDEN

SENIOR VICE PRESIDENT, CHIEF TECHNOLOGY OFFICER AND HEAD OF GROUP FUNCTION TECHNOLOGY

Cloud-native application design

IN THE TELECOM DOMAIN

Cloud-native application design is set to become common practice in the telecom industry in the next few years due to the major efficiency gains that it can provide, particularly in terms of speeding up software upgrades and releases.

HENRIK SAAVEDRA PERSSON,
HOSSEIN KASSAEI

The cloud-native paradigm is driving the transformation of virtual network functions into cloud-native applications (CNAs) that can be commercialized and offered according to either as-a-service (aaS) or as-a-product (aaP) models. In either case, the goal is to provide a seamless and secure deployment, monitoring and operations experience by applying a very high degree of automation.

■ To ease the transition to the cloud-native approach, Ericsson has created an application development framework that provides a set of architecture principles, design rules and best practices that guide the fundamental design decisions for all of our CNAs.

Our framework leverages web-scale technology from the Cloud Native Computing Foundation (CNCF) and other open-source projects while taking into consideration the particular challenges of production-grade telecom applications.

The CNCF is an open-source software foundation whose stated purpose is to make cloud-native computing 'universal and sustainable.' It fosters collaboration between the industry's top developers, end users, and vendors, serving as the vendor-neutral home for many of the fastest-growing projects on GitHub, including Kubernetes, Prometheus and Envoy. CNCF technology has played an important role in our efforts to develop and refine our approach to CNA design.

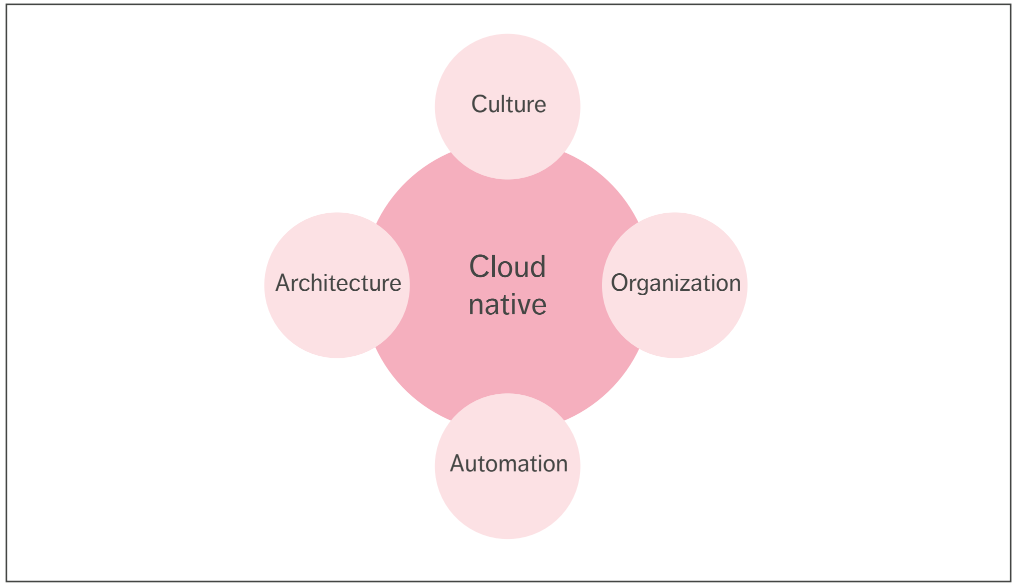

Figure 1 The four pillars of the cloud-native paradigm

Figure 1 illustrates the four pillars of the cloud-native paradigm. Our framework addresses three of them: automation, architecture and culture. Automation is an integral part of the framework, which takes a CI/CD (Continuous Integration, Continuous Delivery) approach to application development and delivery. Architecturally, the framework provides the software assets/components that enable applications to fulfill key design principles [1]. Culturally, it promotes collaboration with the open-source community, as using and contributing to the relevant open-source software projects (typically within CNCF) is at the heart of our implementation strategy.

Our application development framework

Our framework establishes a set of principles for telecom applications based on microservices, containers and state-optimized design. It provides a set of best practices, design rules and guidelines on

Terms and abbreviations

AAP – As-a-Product | **AAS** – As-a-Service | **ACID** – Atomicity, Consistency, Isolation, and Durability | **CAP** – Consistency, Availability and Partition Tolerance | **CAT** – Configuration Assessment Tool | **CI/CD&D** – Continuous Integration, Continuous Delivery and Deployment | **CIS** – The Center for Internet Security | **CNA** – Cloud-native Application | **CNCF** – Cloud Native Computing Foundation | **DR** – Design Rule | **ETSI** – European Telecommunications Standards Institute | **MSA** – Microservice Architecture | **NIST** – National Institute of Standards and Technology | **UI** – User Interface how to build CNAs based on microservice architecture (MSA), as well as guidance on how to deploy, monitor and operate them based on DevOps principles.

With the support of our framework, it is possible to build telecom applications that use CNCF technology through a highly modular architecture and clear separation of concerns. The framework helps us drive alignment across all Ericsson CNAs, ensuring that we address key concerns in a common, generic way. The consistent life-cycle management, operation and maintenance that result from this approach enhance the customer experience. *Figure 2* provides a high-level picture of what the framework offers.

Designing cloud-native applications

Ericsson CNAs are built as a set of loosely coupled (micro)services with well-defined, bounded contexts and individual life cycles. Each microservice is packaged and delivered as one or more containers, independent from other microservices, and provides well-defined and version-controlled application programming interfaces exposed over the network.

To achieve full portability across various infrastructures, CNAs rely on Kubernetes as the choice of container orchestration platform and can be deployed on any certified Kubernetes distribution [2] with a minimum version adhering to the company's security and stability requirements.

All Ericsson CNAs are fully verified on Ericsson Kubernetes distribution. Our CNAs rely on Kubernetes for the automatic placement, auto-scaling, upgrade and auto-healing of individual services. On top of making use of Kubernetes, we also contribute features back to Kubernetes that make it a better fit for telco-grade deployments. IPv6 is just one example of an important area within the telecom domain that has not yet received enough attention within the community.

Observability, security and persistence

Observability is a prerequisite for seamless CNA monitoring and operations. The CNCF landscape [3] includes several very good candidates to help collect,

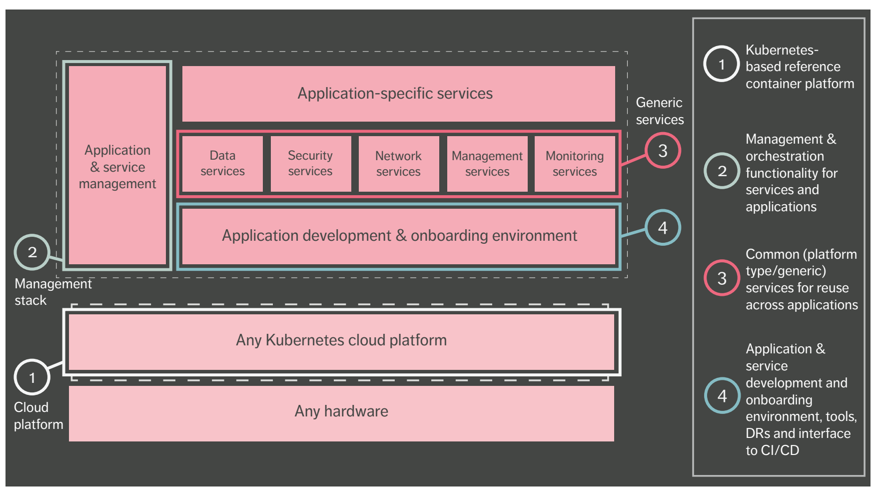

Figure 2 Key components of Ericsson's application development framework

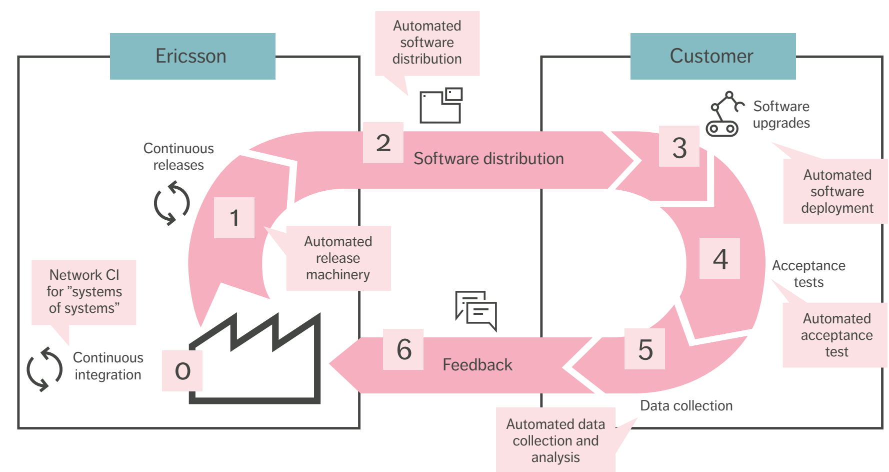

Figure 3 Fully automated CI/CD&D

store and visualize logs, metrics, traces and other data points, such as Prometheus, Fluentd, Elastic Stack, Jaeger and Grafana.

Security is a vital component of cloud-native development. On top of adhering to the best practices and guidelines provided by prominent organizations such as CIS (The Center for Internet Security) and NIST (the National Institute of Standards and Technology), open-source software projects such as Keycloak and HashiCorp Vault can help CNAs deal with storage and provisioning, as well as the handling of identities, certificates and keys.

To break down and implement business logic using stateless microservices, CNAs typically need to rely on stateful backing services to store their data. The type of stateful backing service that is required depends on various factors, such as the type and format of the data (such as structured or unstructured), the amount of data, the intensity of read and write operations, CAP and ACID properties, and so on. A multitude of open-source projects aims to address these needs, including database technologies such as PostgreSQL, MariaDB, Couchbase, Redis, MongoDB, Cassandra, MySQL and Hadoop.

The design philosophy behind Ericsson CNAs is to use polyglot persistence [4] while taking into account the total footprint and avoiding technology sprawl. Achieving the latter requires the identification of the most important properties that enable classification of database engine types into distinct groups and adopting a slightly opinionated approach in selecting one or a few choices in each group.

Continuous Integration, Continuous Delivery and Deployment

Our framework provides tools, interfaces and design rules that enable microservices to benefit from a fully automated Continuous Integration, Continuous Delivery and Deployment (CI/CD&D) pipeline, as illustrated in *Figure 3*. The pipeline is triggered from the moment code is committed and takes the new "candidate release" through the full cycle of build, verification, packaging and release. The deployment phase will be somewhat different for aaS and aaP models. To get the full benefit from cloud native in an aaP approach, the end-to-end pipeline should be fully automated, connecting Ericsson with the customer to allow continuous feedback from live deployments to development teams.

EXPOSURE IS LIMITED TO THE INTERFACE NEEDED BY THE USER OF THE SERVICE

Software reuse and open-source projects

While the goal of large-scale software reuse and utilization of open-source projects existed well before the emergence of the cloud-native paradigm and MSA, it is much more likely to be achieved now. This is because container technology isolates the different services from each other to a very high degree. Instead of being exposed to all the dependencies of each service, the exposure is limited to the interface needed by the user of the service.

Strict backward-compatibility of the exposed interfaces is required to achieve loose coupling and make it possible for each service to evolve independently. At the same time, semantic versioning on the interfaces creates a common way to communicate the evolution of the interfaces.

The CNCF provides a new starting point when looking for reuse opportunities from a cloud-native and MSA perspective. It adds value by creating a structure, choosing relevant open-source projects, and ensuring overall quality and community acceptance. It also provides a comprehensive map of potential realizations for different areas.

Key architectural aspects to consider

Making the right selection between the various CNCF and other open-source projects that are available requires a clear view of the context and the kind of use cases a selected service must support. Having a clearly defined architecture – with set goals, principles, design rules and guidelines – is more important than ever before in this situation because it helps to determine the different service and functional needs.

Following this approach, it is possible to disconnect the definition of which use cases are to be provided for within a particular area (and what additional principles would apply) from a particular realization. The benefit of this is that the architecture itself is not compromised or influenced by the need to support particular use cases (or not). For example, in the case of service mesh, this approach makes it possible to identify the core functional use cases that would add value to the architecture for CNAs, without being influenced by realizations like Istio and Linkerd.

Using identified use cases when considering different potential realizations both within and outside of CNCF enables a direct comparison when looking at the compliance to these use cases. This approach allows for reevaluation of previous selections for whatever reason. From an architecture evolution perspective, it is equally important to update identified use cases as soon as new needs arise, which may in turn lead to a new realization selection.

Separation of concerns

Separation of concerns is an important architectural principle that increases the possibility to reuse CNCF and other open-source projects even for very domain-specific use cases. In this context, separation of concerns means that the internal representation of the data should be separated from external representation. This approach makes it possible to apply cloud-native approaches in areas that were driven by purely proprietary implementations in the past.

Demonstrating the additional benefit and the potentially richer feature set offered by these projects provides an opportunity to influence and evolve expectations within the telecom domain. The ongoing evolution of ONAP (the Open Network Automation Platform) and support for high volume stream data collection is one example of this.

It is rare for out-of-the-box solutions from CNCF and other open-source projects to be sufficient to meet the needs of many telecom-specific use cases related to 3GPP, ETSI and other telecom-defined interfaces. There is, however, an obvious benefit to minimizing in-house development of the additions by building them as modular services on top of an available open-source project, when one exists in the relevant area. Realizations that are relatively generic and provide backward-compatible and version-controlled interfaces are preferred because they make it possible to build extensions in a future-proof way.

An example of this approach would be to choose Prometheus as the performance management infrastructure and use its interfaces to consume metrics and build additional support to expose 3GPP-compliant metric report files.

The question of maturity

Another important factor to consider when selecting an open-source project is maturity. This can be measured in terms of the size of the project's community and the number of releases it has that include backward-compatible changes. Selecting a mature project that is fully compliant with the use cases you have in mind would make it possible to take a passive role in the project, simply using the releases as they become available and focusing on building internal competence. Choosing a less mature project that is not fully compliant with your use cases would require you to adopt an active role in the community to influence both how backward-compatibility is managed and the direction of feature evolution in the future.

In either case, it should be noted that using open-source software is not free – it is important to build up internal knowledge related to the software, especially around the needed use cases. It is not an option to be dependent on the community for all support-related questions, particularly when taking on an active role.

Be aware that, during the project selection process, it will typically only be possible to get a snapshot view of how the project handles the critical issues of backward-compatibility on exposed interfaces and semantic versioning. While a project with a longer history should be able to provide a better picture, it still might not be able to offer a complete one. Kafka is an example of a relatively mature project in which changes that broke backward-compatibility (according to our definition) were announced as a minor update in the release, rather than a major release.

IT IS IMPORTANT TO BUILD UP INTERNAL KNOWLEDGE RELATED TO THE SOFTWARE, ESPECIALLY AROUND THE NEEDED USE CASES

Weighing integration potential against best-of-breed

In several cases, open-source projects have been created with a very different context in mind than the one in which they are later used within a defined architecture – particularly in terms of deployment footprint and characteristics aspects. For example, a project may decide to use one or two specific data stores for persistent data, which, given the bigger picture of the CNA architecture, may not be natural fits.

Sometimes a choice like this is made with an aaS context in mind, where a service is deployed once and reused by everyone, as opposed to an aaP context, where a CNA typically needs to be more self-contained and provide its own instances of all services. One example of this is Harbor, a project that has an opinionated selection of both data store and ingress. In many cases, these sorts of issues can be addressed by configuration or influencing the project, but sometimes they lead to a different selection of realization.

As each open-source project is typically independent, functionality overlap is common, and there can sometimes be contradicting views about how particular problems should be solved. While each project can typically provide value within the scope it was designed for, the broader architectural perspective requires the provision of end-to-end value without compromising defined goals and principles. A key aspect of this is figuring out how different projects can be integrated and used in combination to provide greater value.

In light of this, it is important to keep in mind that even though a specific open-source project may be considered best-of-breed, it might not fit well into the full end-to-end value that the larger architecture intends to provide. In this case, it might make more sense to select a less capable project that is a better fit with the overall architectural goals.

For example, existing best-of-breed projects would not be the right choice to build a cohesive, fully integrated visualization solution to provide a high-level view of microservices and their health status, metrics, logs, distributed traces and other artifacts needed for monitoring in a DevOps team. Bundling best-of-breed standalone user interfaces (UIs) such as Grafana for metrics, Kibana for logs, a Jaeger UI for traces and Kubernetes UI for workload monitoring would result in very poor usability and a sub-optimal experience for Ops teams.

> DIFFERENT PROJECTS CAN BE INTEGRATED AND USED IN COMBINATION TO PROVIDE GREATER VALUE

Shared responsibility for security

Certain aspects of security are expected to be covered by open-source projects, while others remain the full responsibility of the development organization. For instance, when it comes to securing communication, there is typically an aligned approach within both the enterprise and telecom domain. This is centered around TLS and OAuth2, especially looking at the evolution of 3GPP, which is moving to web-scale technology protocols like HTTP/2. When this level of security is not provided by the open-source project, it is typically seen as a valid evolution of the project. In some scenarios, though, due to timing or conflict between free and commercial versions of the project, the mitigation is to deploy a proxy in front of the service to address security aspects – albeit at the cost of introducing additional latency.

The model for using open-source software is to bring in the source code – even if pre-baked container images are provided by some projects – and build container images, including the selection of the base operating system image. Because of this, security hardening must be fully controlled by the team. There are standard testing tools such as CIS-CAT (the CIS configuration assessment tool) that help teams evaluate the quality of the hardening they have performed. Such reports can also be used to assure customers of the overall security compliance.

Key organizational and cultural aspects to consider

Speed is the driving force in the cloud-native paradigm, which means that the goal is to streamline the work process as much as possible. Every task that cannot be automated or cut out inhibits the organization's ability to benefit from the cloud-native approach. Moving to the cloud-native paradigm and making use of MSA is therefore much more than just a technological change in how software is built. A number of important organizational and cultural changes must take place to achieve the benefits.

A cloud-native development organization is often structured around architecture and processes. When the need to create a service has been identified, a small team forms to take full responsibility and accountability for that service across all stages of the software life cycle, according to DevOps principles. This structure stands in stark contrast to the traditional one, in which bigger teams typically work on larger software projects alongside teams with dedicated responsibility for horizontal tasks such as release and compliance handling.

Moving from a more traditional software release cycle of every three to six months to much more frequent releases requires both a higher level of automation of the process and a reduction in the number of activities/tasks needed as part of a release. In short, anything that can be automated must be automated, and a few tasks, such as trade compliance, that cannot be automated must be simplified.

> TO FULLY BENEFIT FROM THE CLOUD-NATIVE PARADIGM, SPEED CANNOT BE COMPROMISED

Building trust and acceptance

Developers do not have full control when they work with open-source code. This can lead to trust issues, particularly in organizations that have a history of working with proprietary implementations. Even when there is a buy-in from developers on the decision to use open-source code, it is often necessary to build trust and acceptance within the organization for the fact that open-source code can do as good a job of fulfilling requirements as code that is developed in-house.

The best way to back up the decision to use open-source code is to demonstrate that there is a solid selection process in place along with clear mapping of architecture use cases. From a business point of view, it is important to emphasize that the cost of developing commoditized software is simply not justifiable: selecting open-source software that meets the criteria to a sufficient degree is preferable to creating similar software from scratch.

Regardless of whether a service is based on open-source code or a proprietary implementation, the team structure and the responsibility must be the same. Constant validation of the open-source project is crucial for early detection of any potential issues relating to backward-compatibility.

In-house talent is crucial to the successful use of open-source code. Internal competence building is as important for open-source based work as it is for proprietary implementations, both for troubleshooting purposes and to understand evolution needs.

Conclusion

Cloud-native application design will soon become the telecom domain's new standard for the development of virtual network functions. Ericsson is well prepared for this transition, thanks to our application development framework, which uses cloud-native principles, microservice architecture and several key enabling technologies such as containers and Kubernetes.

To fully benefit from the cloud-native paradigm, speed cannot be compromised. Technological and architectural changes are not enough – transitioning toward the cloud-native paradigm requires major organizational and cultural changes as well.

Achieving the necessary speed requires smaller team setups with full DevOps responsibility and (full) automation of any step that is required as part of the software CI/CD&D and release process. One of the main organizational challenges is to get the developers to let go of full control and trust the use of open source in areas where in-house development has been used in the past.

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The authors would like to thank Mario Angelic for his contribution to this article.

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Meeting 5G latency requirements

WITH INACTIVE STATE

Reducing the amount of signaling that occurs during state transitions makes it possible to significantly lower both latency and battery consumption – critical requirements for many Internet of Things and 5G use cases, including enhanced mobile broadband.

ICARO LEONARDO DA SILVA, GUNNAR MILDH, PAUL SCHLIWA-BERTLING, MAGNUS STATTIN, ALEXANDER VESELY

Many of the performance improvements in 5G New Radio (NR) that are designed to support new Internet of Things (IoT) use cases such as critical control of remote devices and smart transport [1] are based on lessons learned from research and development on 4G LTE networks. One example of this relates to the transition of wireless devices from a power-saving state where data is not exchanged (idle state) to a connected state optimized for data transmissions (connected state).

■ Studies show that a wireless device's transition from a power-saving (idle) state to a connected state is the most frequent high-layer signaling event in existing 4G LTE networks, occurring about 500-1,000 times a day. The transition comprises an extensive signaling sequence between the device and the network, and between network nodes, which can lead to consumer latency issues and high battery consumption.

The combination of 4G/5G research activities and lessons learned from legacy networks has made it possible to develop solutions that reduce the amount of signaling required at these transitions, thereby lowering both latency and battery consumption significantly. The decreased signaling in the network also results in an increase in overall system capacity.

Ericsson's contributions to the 3GPP standardization of solutions in this area include a new Radio Resource Control (RRC) state model adopted in the standalone version of the 5G NR standard. Improved connection, state and mobility handling are key elements of efficient support for current and future 5G use cases with a large and growing number of devices.

Concept development of the inactive state

Allowing wireless devices to enter a low-power state when they are not transmitting or receiving data has always been an important part of achieving a balance between good communication performance and acceptable battery consumption. For many years, two states – connected and idle – were sufficient to meet most needs.

The development of the inactive state has largely been driven by the growing field of Machine-type Communication (MTC). In most MTC scenarios, the amount of data that wireless devices typically exchange with the network is small and usually not urgent enough to justify the high battery consumption required to handle all the signaling involved in the legacy idle-to-connected transition. To address this issue, Ericsson played a leading role in developing the transition enhancements that were introduced in 4G LTE Rel-13, in which two new procedures were standardized: suspend and resume.

In the suspend procedure, the user equipment (UE) – the 3GPP name for wireless devices – stores its radio configuration and security parameters when it transitions from connected to idle. Then, when it needs to connect to the network again (due to some uplink (UL) data being available to transmit, for example) the UE triggers the resume procedure. This involves restoring the previously stored configuration and resuming the connection without the need for extensive signaling with the core network (CN) or having to reestablish security, for example. The resume procedure is similar to the sleep state of a computer, which enables work to be paused and resumed later without repeating tedious start-up procedures.

In parallel with the 4G LTE work that was completed in 2015, Ericsson was also working on the 5G concept, which included challenging latency requirements and providing support for a variety of new and emerging use cases. Without the constraint to comply with an existing state model, it was possible to further optimize the suspend/resume solution in 5G NR by introducing a new state known as inactive. The key benefits of the inactive state are that it significantly reduces latency and minimizes the battery consumption of both smartphones and MTC devices.

In the latter half of 2015, we began to promote the inactive state externally in the context of the Ericsson-led 5G-PPP European project METIS-II, the main 5G pre-standards project [2]. The main goal of the project was to facilitate research discussions with industry players (UE vendors, network vendors, network operators, academic partners and so on) about technical components to bring to the 3GPP during the 5G standardization work.

Key terms

Connected state – The UE is actively involved in sending or receiving data or signaling. Mobility is controlled by the RAN.

Idle state – The UE is in a power-saving state and is known at tracking-area level in the CN.

Inactive state – The UE is in a power-saving state and is known on RNA level in the RAN. Transition to the connected state is optimized.

	UE RAN states →			
System functions ↓	Idle	Connected	Rel-13 suspend	5G inactive
Mobility management	CN	RAN	CN	RAN
Paging trigger	CN	n/a	CN	RAN
UE configuration data storage	CN	CN and RAN	CN and RAN	CN and RAN
UP contexts in RAN	No	Yes	Yes, but DL packets not sent to RAN	Yes, DL packets sent to RAN

Figure 1 Comparison of the allocation of system functions

Figure 1 shows the allocation of basic system functions in diverse UE states, highlighting the evolution from Rel-13 suspend to 5G inactive.

5G NR inactive state procedures

In 2016, it was agreed that the inactive state would be introduced in 5G NR [3], and the specifications were finalized and approved in December 2018 [4, 5]. The most notable enhancements are the suspend and resume procedures, as well as RAN-based location management and RAN paging for UEs in the inactive state. In the suspend procedure, both the UE and the RAN store information about the UE transition from connected to inactive, along with the UE radio protocol configuration. The resume procedure optimizes the transition from inactive to connected by restoring the UE radio protocol configuration. RAN-based location management and RAN paging make it possible for UEs in the inactive state to move around in an area without notifying the network.

Suspend

The main principle of the inactive state is that the UE is able to return to the connected state as quickly and efficiently as possible. When the UE transitions to inactive, both the UE and the RAN store all the information necessary to quickly resume the connection. The message that transitions the UE to inactive state contains a set of parameters used for inactive state operation, such as a RAN Notification Area (RNA) within which the UE is allowed to move without notifying the network. Further, it includes parameters used for secure transition back to the connected state, such as a UE identifier and security information needed to support encrypted resume messages.

Resume

An inactive UE may initiate a resume procedure when there is a need to transmit data or signaling, for example. In this case, the UE transmits an RRC resume request that includes the UE identifier (provided by the serving node to identify the UE's configuration repository) and a security token to verify the legitimacy of the resume request.

Studies of 4G networks show that, in most cases, UEs that leave the power-saving state return to the same RAN node they were previously served by. If, however, the UE resumes in a cell served by a different RAN node, that target node will retrieve the UE configuration from the serving node based on the UE identifier.

After the UE configuration is successfully retrieved, the target node resumes the stored configuration at the UE and applies any necessary modifications, such as the configuration of measurements and the addition or removal of bearers. The respective RRC resume message is integrity protected and encrypted using the security context stored in the network and the UE.

As illustrated on the right side of *Figure 2*, the resume procedure reduces the number of RRC messages exchanged over the radio interface between the UE and the RAN to three (down from seven for idle state). RRC resume also has the possibility of using efficient delta signaling – in which only changed parameters are signaled – to restore the configuration of a UE in the inactive state. This option is not possible for UEs in the idle state.

The reduction in RRC signaling significantly lowers the access latency experienced by UEs, which leads to more responsive end-user service and the ability to support new use cases. It also reduces the power consumption for devices such as battery-powered sensors that only send small, infrequent reports (often less than 100 bytes of data).

RAN-based location management and RAN paging

The transition from the connected to the inactive state is designed to be invisible to the 5G CN. As a result, even when the UE is in the inactive state, the 5G CN treats it as though it were in the connected state – that is, the UE-associated signaling and user-data connection between the 5G CN and the RAN continues. Mobile-terminated signaling and user-plane (UP) data is sent from the CN to the RAN node currently serving the UE.

When the serving RAN node receives signaling or data for a UE in the inactive state, it initiates RAN paging. The paging is performed in an RNA that consists of one or more cells and was assigned to the

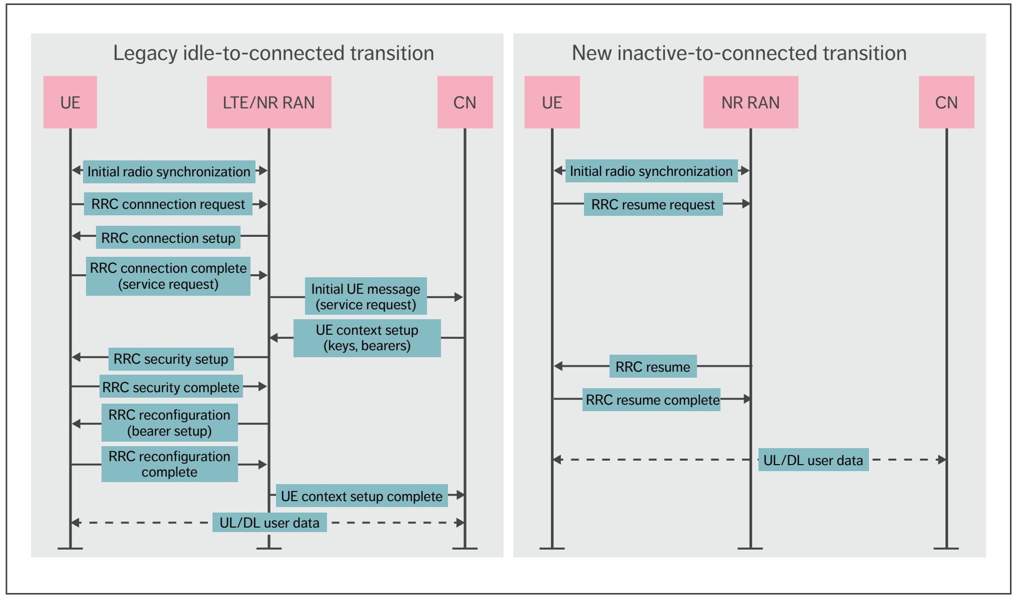

Figure 2 Comparison of signaling involved in legacy idle-to-connected transition (left) versus inactive-to-connected transition (right)

UE when it was ordered to enter the inactive state. When a UE in the inactive state moves to a cell that is not part of its currently assigned RNA, the UE performs a location-update procedure that enables the RAN to update the assigned RNA to the UE.

As in earlier cellular systems, there is a trade-off between the paging load and the amount of location-update signaling. Larger paging areas have more paging load but less location update signaling than smaller paging areas.

Key technology aspects

The most notable technology aspects within the NR inactive state concept adopted by 3GPP are support for encrypted response messages, smart RAN paging, RAN architecture support and fallback to legacy procedure.

Support for encrypted response messages

One of the main components driven by Ericsson in the NR inactive state concept adopted in 3GPP is the ability to encrypt the response message (resume or suspend/release) from the network. This differs from the 4G LTE resume concept adopted in Rel-13, where this message is integrity protected, but sent unencrypted.

To enable the encryption capability, the 3GPP adopted a solution proposed by Ericsson in which the network provides the UE with a security parameter in the release message to the inactive state. The UE uses the security parameter to calculate a new security key to be used when it resumes.

The ability to encrypt the resume response message in 5G NR is advantageous because it makes it possible to use a single, secure message to:

- Reconfigure any parameter in the UE when transitioning to the connected state.
- Release the UE to the idle state (the release message could also include redirection information to another frequency or radio access technology that could be used for voice fallback to LTE, for example).
- Resuspend the UE to the inactive state when it is performing a location-update procedure, for example, so that it only consumes two messages in total (request and response).

Smart RAN paging

For any UE in the connected state, the RAN node receives paging assistance information related to potential paging triggers, such as QoS flows or signaling, from the 5G CN. This information, in combination with other information that the RAN has about the UE, can be used by the RAN node to select and apply a smart paging strategy that is aligned with the characteristics and requirements of the UE and paging-triggering services.

For example, the RAN can configure UE-specific RNAs that make it possible to reduce the total signaling load by configuring small RNAs for stationary UEs (optimized for low paging load) and larger RNAs for moving UEs (optimized for low location update signaling load).

Terms and abbreviations

AMF – Access and Mobility Function | **CA** – Carrier Aggregation | **CN** – Core Network | **CP** – Control Plane | **CU** – Central Unit | **DC** – Dual Connectivity | **DL** – Downlink | **DRX** – Discontinuous Reception | **DU** – Distributed Unit | **EDT** – Early Data Transmission | **E-UTRA** – Evolved Universal Terrestrial Radio Access | **GNB** – GNodeB | **IoT** – Internet of Things | **MAC** – Medium Access Control (protocol) | **MTC** – Machine-type Communication | **NB-IoT** – Narrowband Internet of Things | **NG-RAN** – Next-Generation RAN| **NR** – New Radio | **PPP** – Public-private Partnership | **RNA** – RAN Notification Area | **RRC** – Radio Resource Control | **UE** – User Equipment | **UL** – Uplink | **UP** – User Plane | **UPF** – User-plane Function

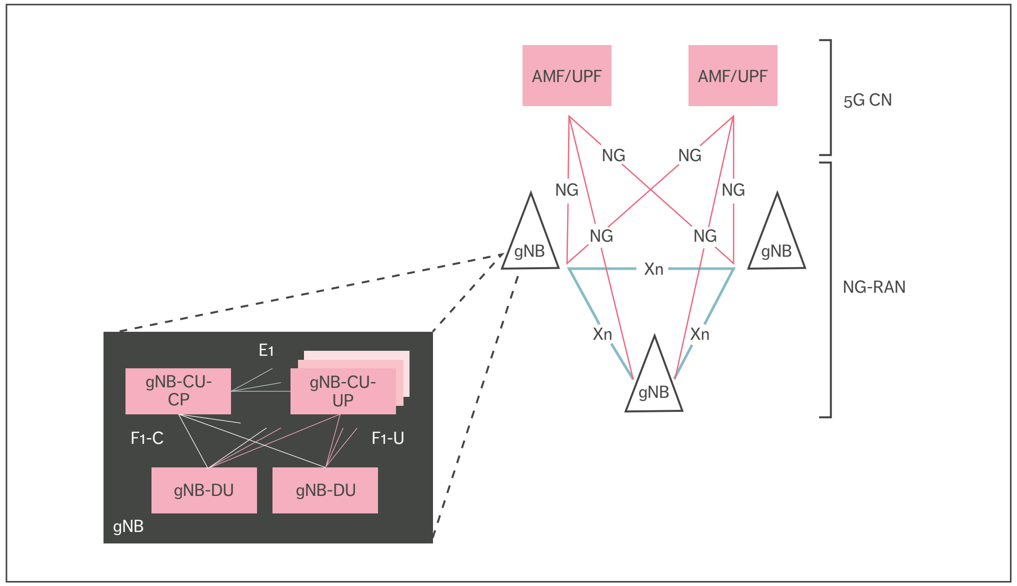

Figure 3 NG-RAN architecture

RAN architecture support

The RAN of the 5G system – known as next-generation RAN (NG-RAN) – consists of RAN nodes that serve either Evolved Universal Terrestrial Radio Access (E-UTRA) or NR cells. The bottom left corner of *Figure 3* illustrates how RAN nodes (gNBs) that serve NR cells can be split into central units (CUs) or distributed units (DUs). A DU hosts functions related to lower radio protocol layers, while a CU hosts functions related to higher radio protocol layers (RRC and Service Data Adaptation Protocol/Packet Data Convergence Protocol). Several DUs are connected to their serving CU nodes via the F1 interface, while RAN nodes may be interconnected by means of the Xn interface. A CU may be further split into a control plane (CP) part (CU-CP) and several UP parts (CU-UP).

The functional decomposition of NG-RAN nodes serves a multitude of different deployment options, including those where a CU is deployed to serve a large number of DUs corresponding to a large serving area. For example, the CU would be able to very efficiently control UE mobility while minimizing signaling traffic toward the 5G CN and between RAN nodes.

With regard to the inactive state, the CU would be able to control tasks such as the assignment of the UE's RNA based on the UE's mobility behavior and certain RAN topology knowledge. Based on Ericsson's proposal, both the CP and UP resources can remain configured in the CU when the UE is in the inactive state. The benefit of this is a reduction in processing and signaling if the UE returns to the same CU, which is highly likely in this type of deployment.

Fallback to legacy procedure

If, for any reason, the UE and the RAN end up in an unsynchronized state and the resume procedure fails, the UE will automatically switch over to the legacy idle-to-connected transition procedure that involves CN signaling. This solution could also be useful if the RAN is unable to retrieve the UE configuration or the UE configuration has been actively discarded.

THE CN IS ABLE TO MAKE CONTACT WITH INACTIVE UEs IF THE RAN CONFIGURATION IS LOST OR DISCARDED

In these situations, the network will respond with an RRC connection setup message instead of an RRC resume message when the UE sends the RRC resume request. When the UE receives the setup message, it will discard the old RAN-related UE configuration and proceed according to the legacy idle-to-connected procedure.

UEs in the inactive state will listen for both RAN- and CN-triggered paging, so that the CN is able to make contact with inactive UEs if the RAN configuration is lost or discarded. This capability is also useful if a UE has been out of radio coverage and missed RAN paging, resulting in the RAN node releasing the UE configuration. To reduce UE battery consumption, the solution provides a mechanism for the RAN and CN to coordinate the paging occasions, so that the UE only needs to wake up once to listen to both.

Future enhancements

While the essential components for the inactive state are supported in Rel-15, there is an opportunity for further enhancements of the NR standard in later releases. There are several use cases and scenarios that would benefit from enhancements to the applicability and efficiency of the inactive state, particularly in the areas of early data transmission (EDT), early measurements and long discontinuous reception (DRX).

Early data transmission

To meet the more stringent requirements of future 5G use cases, it will soon be necessary to reduce UL latency even further. EDT is a feature that would allow opportunistic data transmission to commence during the connection resume procedure. With the resume procedure as specified in Rel-15, connection resume procedures are completed before data transmission can start. With EDT, data transfer can begin in parallel with transmission of the resume request message in the UL and the resume message in the downlink (DL). Security and radio bearers are resumed before submitting the resume request message to lower layers,which allows multiplexing of data with signaling in the Medium Access Control (MAC) layer.

EDT has already been introduced in LTE-M and Narrowband-IoT (NB-IoT), where traffic is expected to comprise the transmission of small amounts of data and one of the primary objectives is long UE battery life. For use cases where traffic consists of only one UL and/or one DL data packet, EDT improves energy efficiency by enabling the network to release the UE to the inactive state without the need for intermediate resume and resume complete messages.

Early measurements

NR Rel-15 already supports the aggregating of multiple carriers for higher data throughput using either carrier aggregation (CA) or dual connectivity (DC). When transitioning from the inactive or idle state to connected state, however, the UE only has access through one carrier. Faster setup of CA or DC would make it possible to further reduce the session setup latency. However, the usefulness of CA and DC depends on the network understanding of the radio environment.

Early measurement reporting is a feature currently being standardized in Rel-16 to improve the setup of CA and DC by enhancing NR to support early radio measurement reports during the transition from the inactive to connected state – that is, in parallel with the resume complete message. This would be possible in NR because when the UE is suspended, it receives the security parameters needed to encrypt the sensitive measurement report. When security is activated, early measurement reports can be multiplexed with a resume request or multiplexed with a resume complete message (if requested by the RAN in the resume message).

Long discontinuous reception

DRX, a feature that enables the UE to turn off its receiver, is imperative in use cases where device energy efficiency and battery life are important considerations. The longer the transmitter and receiver can be turned off, the more energy the UE can save (the longer the battery life). Long DRX has traditionally been supported in the idle state.

To enjoy the benefits of both signaling reductions and long DRX, it is desirable to extend DRX cycles in the inactive state to the same length as DRX cycles in the idle state. A key aspect of the inactive state, however, is that, from the CN point of view, the UE remains connected and DL data arriving to the CN would normally be forwarded to the RAN node serving the UE. The RAN would buffer the data until the UE is reachable.

With short DRX, the amount of data that would need to be buffered is limited. With very long DRX, however, the buffering requirements in the RAN grow and may exceed what is normally needed, which would become costly. To mitigate this and make use of (already existing) CN buffering capability/capacity, the RAN may indicate to the CN that the UE is not available for DL data while in the inactive state. In this event, the CN will buffer the data and notify the RAN, so that the RAN can inform the CN when the UE becomes available again. It is anticipated that the use of long DRX in the inactive state with buffering offloaded to the CN would further improve battery life, while enabling efficient (re)use of buffering capabilities in the network.

Conclusion

Improved connection, state and mobility handling are key requirements of many current and future 5G use cases, including smart transport and critical control of remote devices. At Ericsson, our 4G/5G research activities and lessons learned from legacy networks have enabled us to identify solutions that significantly lower both latency and battery consumption by reducing the amount of signaling required during state transitions. As a result of this work, the standalone version of the 5G NR standard includes a new Radio Resource Control state model that features a new state called inactive.

The inactive state in 5G NR is a key enabler for emerging use cases that require low latency communication and minimal battery consumption. An additional benefit of the new state is that the decreased processing effort in the network results in an increase in overall system capacity.

Rel-15 includes all the essential components for the inactive state. Future releases should focus on providing applicability and efficiency enhancements, particularly in the areas of early data transmission, early measurements and long discontinuous reception.

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5G-TSN integration meets networking requirements

FOR INDUSTRIAL AUTOMATION

The move toward smart manufacturing creates extra demands on networking technologies – namely ubiquitous and seamless connectivity while meeting the real-time requirements. Today, 5G is good for factories; nevertheless, its integration with Time-Sensitive Networking (TSN) would make smart factories fully connected and empower them to meet all key requirements on industrial communication technology.

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BALÁZS VARGA,
GYÖRGY MIKLÓS,
JOACHIM SACHS

Industrial automation is one of the industry verticals that can benefit substantially from 5G, including, for example, increased flexibility, the reduction of cables and support of new use cases [1]. At the same time, factory automation is going through a transformation due to the fourth industrial revolution (also known as Industry 4.0), and this requires converged networks that support various types of traffic in a single network infrastructure.

■ As it stands, IEEE (Institute of Electrical and Electronics Engineers) 802.1 Time-Sensitive Networking (TSN) is becoming the standard Ethernet-based technology for converged networks of Industry 4.0. It is possible for 5G and TSN to coexist in a factory deployment and address their primary requirements, such as 5G for flexibility and TSN for extremely low latency. Beyond that, 5G and TSN can be integrated to provide solutions to the aforementioned demands of ubiquitous and seamless connectivity with the deterministic QoS required by control applications end to end. Ultimately, integrating these key technologies provides what is needed for smart factories.

5G: adding ultra-reliable low-latency communication

5G has been designed to address enhanced mobile broadband services for consumer devices such as smartphones or tablets, but it has also been tailored for Internet of Things (IoT) communication and connected cyber-physical systems. To this end, two requirement categories have been defined: massive machine-type communication for a large number of connected devices/sensors, and ultra-reliable low-latency communication (URLLC) for connected control systems and critical communication [1] [2]. It is the capabilities of URLLC that make 5G a suitable candidate for wireless deterministic and time-sensitive communication. This is essential for industrial automation, as it can enable the creation of real-time interactive systems, and also for the integration with TSN.

Several features have been introduced to 5G in phase 1 (3GPP Release 15) and phase 2 (3GPP Release 16, to be finalized by March 2020) that reduce the one-way latency and enable the transmission of messages over the radio interface with reliability of up to 99.999 percent, achievable in a controlled environment such as a factory.

> ULTIMATELY, INTEGRATING THESE KEY TECHNOLOGIES PROVIDES WHAT IS NEEDED FOR SMART FACTORIES

5G RAN features

5G RAN [3] with its New Radio (NR) interface includes several functionalities to achieve low latency for selected data flows. NR enables shorter slots in a radio subframe, which benefits low-latency applications. NR also introduces mini-slots, where prioritized transmissions can be started without waiting for slot boundaries, further reducing latency. As part of giving priority and faster radio access to URLLC traffic, NR introduces preemption – where URLLC data transmission can preempt ongoing non-URLLC transmissions. Additionally, NR applies very fast processing, enabling retransmissions even within short latency bounds.

Definition of key terms

Smart factories are being developed as part of the fourth industrial revolution. They require ubiquitous connectivity among and from the devices to the cloud through a fully converged network, supporting various types of traffic in a single network infrastructure, which also includes mobile network segments integrated into the network.

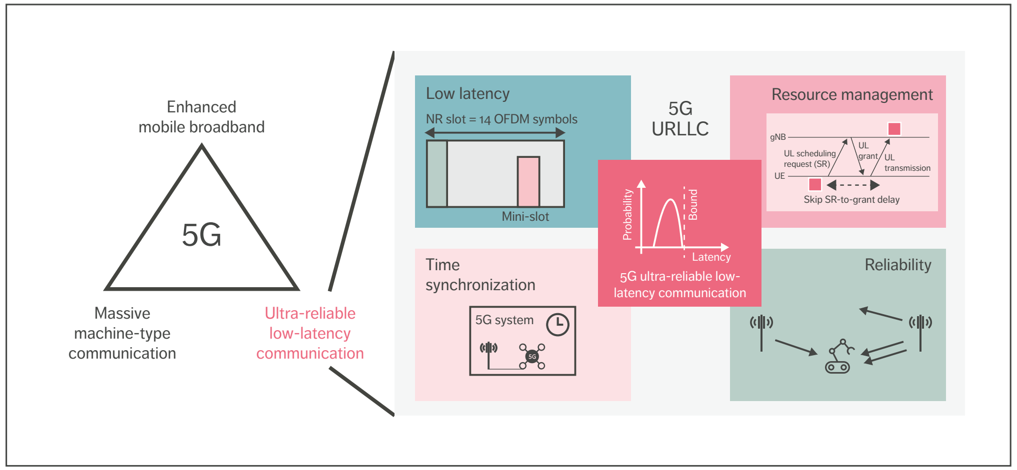

Figure 1 5G URLLC overview

5G defines extra-robust transmission modes for increased reliability for both data and control radio channels. Reliability is further improved by various techniques, such as multi-antenna transmission, the use of multiple carriers and packet duplication over independent radio links.

Time synchronization is embedded into the 5G cellular radio systems as an essential part of their operation, which has already been common practice for earlier cellular network generations. The radio network components themselves are also time synchronized, for instance, through the precision time protocol telecom profile [4]. This is a good basis to provide synchronization for time-critical applications.

Figure 1 illustrates URLLC features. It shows that 5G uses time synchronization for its own operations, as well as the multiple antennas and radio channels that provide reliability. 5G brings in redefined schemes for low latency and resource management, which can be combined to provide ultra-reliability and low latency.

Besides the 5G RAN features, the 5G system (5GS) also provides solutions in the core network (CN) for Ethernet networking and URLLC. The 5G CN supports native Ethernet protocol data unit (PDU) sessions. 5G assists the establishment of redundant user plane paths through the 5GS, including RAN, the CN and the transport network. The 5GS also allows for a redundant user plane separately between the RAN and CN nodes, as well as between the UE and the RAN nodes.

Time-Sensitive Networking for converged networks

TSN provides guaranteed data delivery in a guaranteed time window; that is, bounded low latency, low-delay variation and extremely low data loss, as illustrated in Figure 2. TSN supports various kinds of applications having different QoS requirements: from time- and/or mission-critical data traffic, for example, closed-loop control, to best-effort traffic over a single standard Ethernet network infrastructure; in other words, through a converged network. As a result, TSN is an enabler of Industry 4.0 by providing flexible data access and full connectivity for a smart factory.

Time-Sensitive Networking standards

TSN is a set of open standards specified by IEEE 802.1 [5]. TSN standards are primarily for IEEE Std 802.3 Ethernet, which means they utilize all the benefits of standard Ethernet, such as flexibility, ubiquity and cost savings.

TSN standards can be seen as a toolbox that includes several valuable tools, which can be categorized into four groups: traffic shaping, resource management, time synchronization and reliability, as shown in *Figure 2*. Here, we focus only on the TSN tools that are strong candidates for early TSN deployments in industrial automation.

TSN guarantees the worst-case latency for critical data by various queuing and shaping techniques and by reserving resources for critical traffic. The Scheduled Traffic standard (802.1Qbv) provides time-based traffic shaping. Ethernet frame preemption (802.3br and 802.1Qbu), which can suspend the transmission of a non-critical Ethernet frame, is also beneficial to decrease latency and latency variation of critical traffic.

Resource management basics are defined by the TSN configuration models (802.1Qcc). Centralized Network Configuration (CNC) can be applied to the network devices (bridges), whereas, Centralized User Configuration (CUC) can be applied to user devices (end stations). The fully centralized configuration model follows a software-defined networking (SDN) approach; in other words, the CNC and CUC provide the control plane instead of distributed protocols. In contrast, distributed control protocols are applied in the fully distributed model, where there is no CNC or CUC.

High availability, as a result of ultra-reliability, is provided by Frame Replication and Elimination for Reliability (FRER) (802.1CB) for data flows through a per-packet-level reliability mechanism. This provides reliability by transmitting multiple copies of the same data packets over disjoint paths in the network. Per-Stream Filtering and Policing (802.1Qci) improves reliability by protecting against bandwidth violation, malfunctioning and malicious behavior.

The TSN tool for time synchronization is the

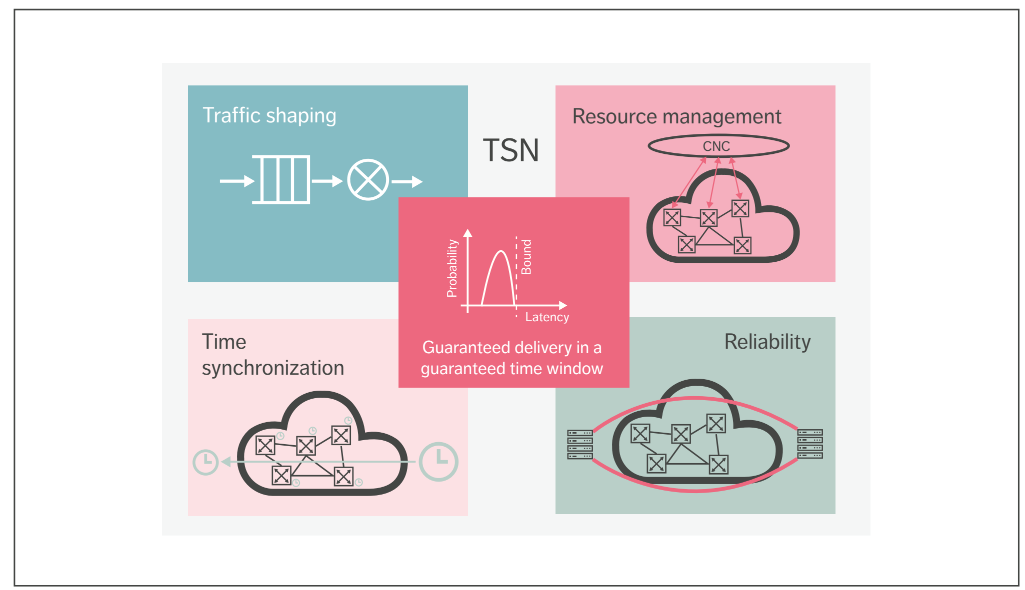

Figure 2 Valuable tools within the TSN toolbox that enable deployments in industrial automation

5G-TSN INTEGRATION FOR INDUSTRIAL AUTOMATION ✱

generalized Precision Time Protocol (gPTP) (802.1AS), which is a profile of the Precision Time Protocol standard (IEEE 1588). The gPTP provides reliable time synchronization, which can be used by other TSN tools, such as Scheduled Traffic (802.1Qbv).

It is important to note that TSN standards are built upon the base IEEE 802.1 bridging standards, some of which have to be supported in TSN deployments as well – including industrial automation.

A special set of TSN standards are the TSN profiles because a profile selects TSN tools and describes their use for a particular use case or vertical.

Time-Sensitive Networking for industrial automation

The IEC/IEEE 60802 profile [6] specifies the application of TSN for industrial automation, and also gives guidelines to what 5G needs to support. IEC/IEEE 60802 provides basis for other standards targeting interoperability in industrial automation. For instance, Open Platform Communications (OPC) Foundation's Field Level Communications [7] initiative aims for one common multi-vendor converged TSN network infrastructure.

The IEC/IEEE 60802 profile will specify multiple classes of devices. There will be at least two classes of devices for both device types – bridges and end stations. One class is feature rich (currently called Class A), and the other class is constrained (currently called Class B), meaning that it supports a smaller set of features. Bridges and end stations belonging to the same class have the same mandatory and optional TSN capabilities.

The Link Layer Discovery Protocol (LLDP) (802.1AB) is mandatory for all device types and classes for the discovery of the network topology and neighbor information.

Time synchronization is also mandatory for all device types and classes. The current target is to support a minimum of three time domains for Class A and a minimum of two time domains for Class B.

Class A devices must support a wide range of TSN functions (such as Scheduled Traffic, Frame Preemption, Per-Stream Filtering and Policing, FRER and TSN configuration), which are optional for Class B devices.

Integrated 5G and Time-Sensitive Networking

5G URLLC capabilities provide a good match to TSN features (as illustrated in Figures 1 and 2). The two key technologies can be combined and integrated to provide deterministic connectivity end to end, such as between input/output (I/O) devices and their controller potentially residing in an edge cloud for industrial automation. The integration includes support for both the necessary base-bridging features and the TSN add-ons.

Terms and abbreviations

5GS – 5G System | **5QI** – 5G QoS Indicator | **AF** – Application Function | **CN** – Core Network | **CNC** – Centralized Network Configuration | **CUC** – Centralized User Configuration | **FRER** – Frame Replication and Elimination for Reliability | **gNB** – Next generation Node B (5G base station) | **gPTP** – Generalized Precision Time Protocol | **I/O** – Input/Output | **IEC** – International Electrotechnical Commission | **IEEE** – Institute of Electrical and Electronics Engineers | **IoT** – Internet of Things | **LLDP** – Link Layer Discovery Protocol | **NR** – New Radio | **OFDM** – Orthogonal Frequency Division Multiplexing | **OPC** – Open Platform Communications | **PCF** – Policy Control Function | **PDU** – Protocol Data Unit | **SDN** – Software-Defined Networking | **TSN** – Time-Sensitive Networking | **TT** – TSN Translator | **UE** – User Equipment | **UL** – Uplink | **UPF** – User Plane Function | **URLLC** – Ultra-Reliable Low-Latency Communication

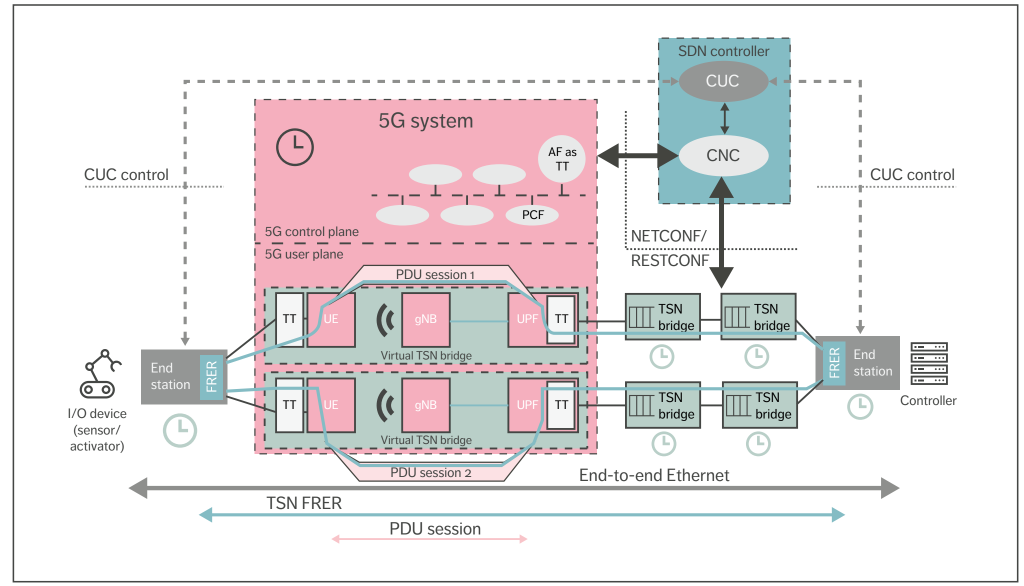

Figure 3 5GS integrated with TSN providing end-to-end deterministic connectivity

Figure 3 illustrates the 5G-TSN integration, including each TSN component shown in Figure 2. It shows the fully centralized configuration model, which is the only configuration model supported in 5G phase 2 (3GPP Release 16).

The 5GS appears from the rest of the network as a set of TSN bridges – one virtual bridge per User Plane Function (UPF) as shown in the figure. The 5GS includes TSN Translator (TT) functionality for the adaptation of the 5GS to the TSN domain, both for the user plane and the control plane, hiding the 5GS internal procedures from the TSN bridged network.

The 5GS provides TSN bridge ingress and egress port operations through the TT functionality. For instance, the TTs support hold and forward functionality for de-jittering. The figure illustrates functionalities using an example of two user equipments (UEs) with two PDU sessions supporting two correlated TSN streams for redundancy. But a deployment may only include one physical UE with two PDU sessions using dual-connectivity in RAN. The figure illustrates the case when the 5GS connects an end station to a bridged network; however, the 5GS may also interconnect bridges.

The support for base bridging features described here is applicable whether the 5G virtual bridges are Class A or Class B capable. The 5GS has to support the LLDP features needed for the control and management of an industrial network, such as for the discovery of the topology and the features of the 5G virtual bridges. The 5GS also needs to adapt to the loop prevention method applied in the bridged network, which may be fully SDN controlled without any distributed protocol other than LLDP.

5G supporting Time-Sensitive Networking

Ultra-reliability can be provided end to end by the application of FRER over both the TSN and 5G domains. This requires disjoint paths between the FRER end points over both domains, as illustrated in Figure 3.

5G-TSN INTEGRATION FOR INDUSTRIAL AUTOMATION ✱

THE 5G-TSN INTEGRATION IS A KEY TOPIC OF IMPORTANCE AT ERICSSON

A 5G UE can be configured to establish two PDU sessions that are redundant in the user plane over the 5G network [2]. The 3GPP mechanism involves the appropriate selection of CN and RAN nodes (UPFs and 5G base stations (gNBs)), so that the user plane paths of the two PDU sessions are disjoint. The RAN can provide the disjoint user plane paths based on the use of the dual-connectivity feature, where a single UE can send and receive data over the air interface through two RAN nodes.

The additional redundancy – including UE redundancy – is possible for devices that are equipped with multiple UEs. The FRER end points are outside of the 5GS, which means that 5G does not need to specify FRER functionality itself. Also, the logical architecture does not limit the implementation options, which include the same physical device implementing end station and UE.

Requirements of a TSN stream can be fulfilled only when resource management allocates the network resources for each hop along the whole path. In line with TSN configuration (802.1Qcc), this is achieved through interactions between the 5GS and CNC (see Figure 3). The interface between the 5GS and the CNC allows for the CNC to learn the characteristics of the 5G virtual bridge, and for the 5GS to establish connections with specific parameters based on the information received from the CNC.

Bounded latency requires deterministic delay from 5G as well as QoS alignment between the TSN and 5G domains. Note that 5G can provide a direct wireless hop between components that would otherwise be connected via several hops in a traditional industrial wireline network. Ultimately, the most important factor is that 5G can provide deterministic latency, which the CNC can discover together with TSN features supported by the 5GS.

For instance, if a 5G virtual bridge acts as a Class A TSN bridge, then the 5GS emulates time-controlled packet transmission in line with Scheduled Traffic (802.1Qbv). For the 5G control plane, the TT in the application function (AF) of the 5GS receives the transmission time information of the TSN traffic classes from the CNC. In the 5G user plane, the TT at the UE and the TT at the UPF can regulate the time-based packet transmission accordingly. TT internal details are not specified by 3GPP and are left for implementation. For example, a play-out (de-jitter) buffer per traffic class is a possible solution. The different TSN traffic classes are mapped to different 5G QoS Indicators (5QIs) in the AF and the Policy Control Function (PCF) as part of the QoS alignment between the two domains, and the different 5QIs are treated according to their QoS requirements.

Time synchronization

Time synchronization is a key component in all cellular networks (illustrated by the black 5GS clock in Figure 3). Providing time synchronization in a 5G-TSN combined industrial deployment brings in new aspects. In most cases, end devices need time reference regardless of whether it is used by TSN bridges for their internal operations. Bridges also require time reference if they use a TSN feature that is based on time, such as Scheduled Traffic (802.1Qbv). The green clocks in Figure 3 illustrate a case when both bridges and end stations are time synchronized.

As gPTP is the default time synchronization solution for TSN-based industrial automation, the 5GS needs to interwork with the gPTP of the connected TSN network. The 5GS may act as a virtual gPTP time-aware system and support the forwarding of gPTP time synchronization information between end stations and bridges through the 5G user plane TTs. These account for the residence time of the 5GS in the time synchronization procedure. One special option is when the 5GS clock acts as a grandmaster and provides the time reference not only within the 5GS, but also to the rest of the devices in the deployment, including connected TSN bridges and end stations.

Overall, 5G standardization has addressed the key aspects needed for 5G-TSN integration.

Conclusion

Together, 5G and Time-Sensitive Networking (TSN) can meet the demanding networking requirements of Industry 4.0. The 5G-TSN integration is a key topic of importance at Ericsson, and we see that the combination of 5G and TSN is perfect for smart factories, given the features provided for ultra-reliability and low latency. That said, a certain level of integration of the two technologies is needed to provide an end-to-end Ethernet connectivity to meet the industrial requirements.

Integrated time synchronization via wireless 5G and wired TSN domains provides a common reference time for industrial end points. 5G is also integrated with the given TSN tool used in a particular deployment to provide bounded low latency. The disjoint forwarding paths of the 5G and TSN segments are aligned to provide end-to-end ultra-reliability and high availability.

The first step of control plane integration is being carried out for a software-defined networking-based approach (the fully centralized model of TSN). Fundamentally, 5G and TSN include the key technology components required for combined deployment in industrial automation and high availability.

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Further reading

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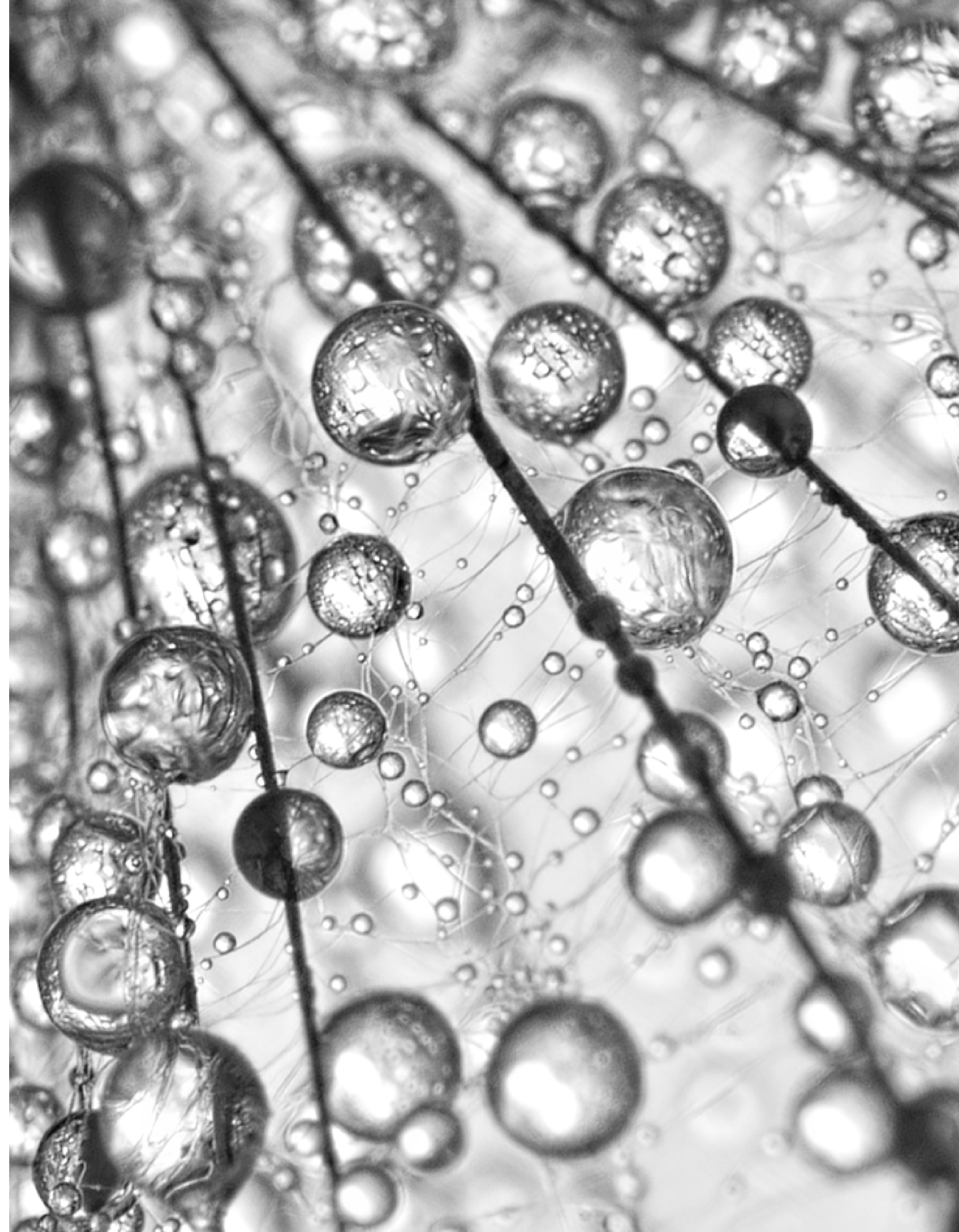

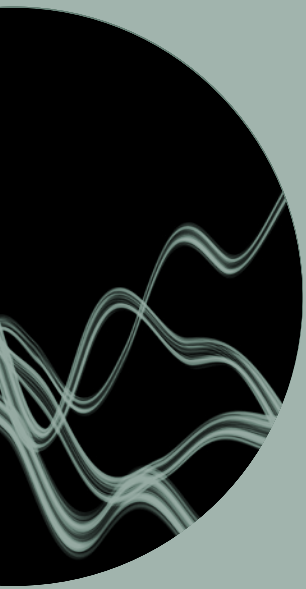

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